



Historical perspective

Superhydrophilicity and strong salt-affinity: Zwitterionic polymer grafted surfaces with significant potentials particularly in biological systems

Dongxiang Li^{a,*}, Qianling Wei^a, Chunxing Wu^a, Xiaofang Zhang^a, Qihao Xue^a, Taoran Zheng^a, Meiwen Cao^b^a Key Laboratory of Optic-electro Sensing and Analytical Chemistry of Life Science (Ministry of Education), Lab of Colloids and Functional Nanostructures, College of Chemistry and Molecular Engineering, Qingdao University of Science and Technology, Qingdao 266042, China^b State Key Laboratory of Heavy Oil Processing and Centre for Bioengineering and Biotechnology, College of Chemical Engineering, China University of Petroleum (East China), Qingdao 266580, China

ARTICLE INFO

Article history:

11 March 2020

Available online 16 March 2020

Keywords:

Zwitterionic polymer

Surface grafting

Superhydrophilic

Salt-responsive

Antifouling

Biocompatible

ABSTRACT

In recent years, zwitterionic polymers have been frequently reported to modify various surfaces to enhance hydrophilicity, antifouling and antibacterial properties, which show significant potentials particularly in biological systems. This review focuses on the fabrication, properties and various applications of zwitterionic polymer grafted surfaces. The “graft-from” and “graft-to” strategies, surface grafting copolymerization and post zwitterionization methods were adopted to graft lots type of the zwitterionic polymers on different inorganic/organic surfaces. The inherent hydrophilicity and salt affinity of the zwitterionic polymers endow the modified surfaces with antifouling, antibacterial and lubricating properties, thus the obtained zwitterionic surfaces show potential applications in biosystems. The zwitterionic polymer grafted membranes or stationary phases can effectively separate plasma, water/oil, ions, biomolecules and polar substrates. The nanomedicines with zwitterionic polymer shells have “stealth” effect in the delivery of encapsulated drugs, siRNA or therapeutic proteins. Moreover, the zwitterionic surfaces can be utilized as wound dressing, self-healing or oil extraction materials. The zwitterionic surfaces are expected as excellent support materials for biosensors, they are facing the severe challenges in the surface protection of marine facilities, and the dense ion pair layers may take unexpected role in shielding the grafted surfaces from strong electromagnetic field.

© 2020 Elsevier B.V. All rights reserved.

Contents

1.	Introduction	2
2.	Classification of zwitterionic polymers	3
2.1.	Basic categories	3
2.2.	Detail of betaines	3
3.	Fabrication methods	3
3.1.	“Graft-from”	3
3.1.1.	ATRP	3
3.1.2.	RAFT polymerization	6
3.1.3.	Physisorbed free radical polymerization	6
3.1.4.	Ring-opening metathesis polymerization	6
3.1.5.	Photo-, thermal- and Plasma- induced polymerization	6
3.2.	“Graft-to”	6
3.2.1.	Physical adsorption	6
3.2.2.	Biomimetic bonding	6
3.2.3.	Chemical bonding	6
3.3.	Surface grafting copolymerization	6
3.4.	Post zwitterionization	7
4.	Hydrophilicity, salt affinity and other properties	7
4.1.	Microscopic analysis	7

* Corresponding author at: College of Chemistry and Molecular Engineering, Qingdao University of Science and Technology, Zhengzhou Road No. 53, Qingdao 266042, China.
E-mail addresses: lidx@qust.edu.cn, lidx@ccas.ac.cn (D. Li).

4.2.	Hydrophilicity	7
4.3.	Salt responsive mechanism and salt affinity	8
4.4.	Antifouling properties.	8
4.5.	Lubricating properties.	9
5.	Applications	10
5.1.	Grafted membranes for separation	10
5.1.1.	Plasma separation	10
5.1.2.	Water separation.	10
5.1.3.	Salt separation	11
5.2.	Grafted stationary phases for HILIC column separation	11
5.3.	Grafted solid surfaces	12
5.3.1.	Organic surfaces	12
5.3.2.	Inorganic surfaces	12
5.3.3.	Implanted material surfaces	13
5.4.	Nanomedicines.	13
5.4.1.	Zwitterionic therapeutic proteins	13
5.4.2.	Grafted solid nanoparticles	14
5.5.	Miscellaneous cases.	14
6.	Summary and outlook	14
	Acknowledgements	15
	References.	15

Nomenclature

polyC ₂ BMA	poly[2-(methacryloyloxy)ethyl 1-carboxymethyl dimethyl ammonium]
polyC ₃ BMA	poly[2-(methacryloyloxy)ethyl 2-carboxylethyl dimethyl ammonium]
polyC ₂ BMAA	poly[3-(methacrylamide)propyl 1-carboxymethyl dimethyl ammonium]
polyC ₃ BMAA	poly[3-(methacrylamide)propyl 2-carboxylethyl dimethyl ammonium]
polyC ₃ BAA	poly[3-(acrylamide)propyl 2-carboxylethyl dimethyl ammonium]
polyCBOH	poly[2-(2-hydroxy-3-methacryloyloxy propyl dimethyl ammonio) acetate]
polyDVBAMS	poly[dimethyl 4-vinylbenzyl 1-ammonio methanesulfonate]
polyDVBAPS	poly[dimethyl 4-vinylbenzyl 3-ammonio propanesulfonate]
polyDVBABS	poly[dimethyl 4-vinylbenzyl 4-ammonio butanesulfonate]
polyMPC	poly(2-methacryloyloxyethyl phosphorylcholine)
polyNCIPS	poly(5-norbornene-2-ylcarboxyethyl-1-imidazolium propane sulfonate)
polySBMA	poly[2-(methacryloyloxy)ethyl 3-sulfopropyl dimethyl ammonium]
polySBAAE	poly[2-(acrylamide)ethyl 3-sulfopropyl dimethyl ammonium]
polySBMAAP	poly[3-(methacrylamide)propyl 3-sulfopropyl dimethyl ammonium]
polySBAAP	poly[3-(acrylamide)propyl 3-sulfopropyl dimethyl ammonium]
polySVBP	poly[4-(2-sulfethyl) 1-(4-vinylbenzyl) pyridinium]
polyVBIPS	poly[3-[1-(4-vinylbenzyl)-1-H-imidazol-3-ium-3-yl] propane sulfonate]
polyVPPS	poly(4-vinyl-pyridinium-1-propane sulfonate)
polyDMAEMA	poly[2-(dimethylamino)ethyl methacrylate]
PE	polyethylene
PEG	poly(ethylene glycol)
PEGMA	poly(ethylene glycol methacrylate)
PEI	polyethyleneimine
PES	polyethersulfone
PVDF	poly(vinylidene fluoride)

1. Introduction

Bacterial attachment and subsequent surface-induced reactions on the surface of biomaterials always lead to the formation of biofilms and cause serious problems in human healthcare and industrial applications [1–3]. Surface-induced biological reactions, including non-specific protein adsorption, blood clot formation, and cell/bacterial adhesion, will interfere with device function, lead to failure of implanted devices, and threaten patient life. To our knowledge, many surfaces from metals, inorganic glass, silicon wafer to organic rubber, plastic materials, either planar or curved particle surfaces, can be modified by hydrophilic polymers to adjust their surface behavior [4–6], especially to enhance the biocompatibility for biological applications [7–11]. Therefore, the hydrophilic polymers such as PEG have been used to modify the material surface in the early years in order to overcome these surface-induced biological reactions. However, PEG and its derivatives are reported to be easily destroyed by oxidation in the presence of transition metal ions or oxygen [12–14].

Recent works have demonstrated that zwitterionic compounds with a mixture of anionic and cationic terminal groups, such as betaines and phosphorylcholine, have strong antifouling properties and can solve the above limitations [15–17], and the zwitterionic components can be functionalized with molecular design freedom and chemical diversity [18]. In particular, zwitterionic polymers have proven to be of high hydrophilicity, long-term durability and environmental stability [19,20], they exhibit excellent antibiofouling property and are considered as alternatives to currently used PEG in the biomedical field [21–24], because the ionic solvation of the zwitterionic polymers produces high retention of water and forms a strong hydration layer around polymer chains to effectively prevent non-specific adsorption of proteins [25,26]. So that, the excellent antifouling surfaces with zwitterionic polymer brushes are expected to avoid the surface-induced biological reactions, thereby they are becoming a research hotspot in the interdisciplinary field of surface chemistry and polymer materials. Nowadays, fabricating zwitterionic polymer brushes on a variety of surfaces have proved to be a very versatile and effective strategy for surface modification in many from-bench-to-bedside applications including drug delivery, artificial organ implants, biomaterials and biosensors [14,27–29]. For example, certain micropatterns of polySBMAAP and PEGMA on silicon oxide surfaces were used to investigate the long-term stability of patterned polymer brushes, in which the highly improved non-biofouling ability and hydrophilicity of polySBMAAP brushes were thought to contribute to the observed effective resistance against cell

adhesion for a longer period time than PEGMA brushes of the same thickness [30].

Therefore, a large number of literatures were recently published on the zwitterionic polymer grafted surfaces for fascinating applications in many fields [31–35]. Herein, this review mainly focuses on the recent advances of zwitterionic polymer grafted surfaces from fabrication methods, hydrophilicity, salt-affinity, antifouling and lubricating properties to a wide variation of applications such as substrate separations, implanting materials and nanomedicines in biological systems. This contribution is expected to provide a more comprehensive understanding of the zwitterionic polymer grafted surfaces.

2. Classification of zwitterionic polymers

2.1. Basic categories

As shown in Fig. 1, zwitterionic polymers are always composed of a backbone main chain and many side chains with positive and negative ions. Based on the differences of zwitterionic side chains, the zwitterionic polymers can be roughly divided into two categories. The first category is betaine-like monomers, which carry both an anionic group and a cationic group together in one unit (Fig. 1A) [36–38]. The betaine segments include phosphorylcholine (PC) [39–73], sulfobetaine (SB) [66–69,72–124], carboxybetaine (CB) [22,108–110,124–141] and other types [142–145]; meanwhile, the length of carbon spacer between anion and cation is variable to adjust the properties of the polymers [122]. In the second category, the zwitterionic polymers are factually polyampholytes, that is, special block copolymers which carry 1:1 positive and negative charges on different monomer units (Fig. 1B) [146–149]. The uniformity of charge distribution and charge neutrality of polyampholytes can be mostly achieved by a homogeneous mixture of 1:1 oppositely charged comonomers in copolymerization reaction [18,146]. These opposite charges in the polyampholytes thereby are evenly distributed throughout the polymer chain, which makes the polyampholyte molecules zero-charged [32]. On the other hand, the backbone main chain of these polymers is usually polymerized from carbon double-bond monomers of methacrylate, methacrylamide, acrylamide, vinylbenzyl or other derivatives.

2.2. Detail of betaines

Most of the published works have focused on the first category, betaine-like zwitterionic monomers, of which the chemical structure, name, abbreviation and references are listed in Table 1. All reported phosphorylcholine monomers are methacryloyloxyethyl phosphorylcholine (MPC) [39–73], while sulfobetaine monomers include 2-(methacryloyloxy)ethyl 3-sulfopropyl dimethyl ammonium (SBMA) [69,72–116], 2-(acrylamide)ethyl 3-sulfopropyl dimethyl ammonium (SBAAE) [120], 3-(methacrylamide)propyl 3-sulfopropyl dimethyl ammonium (SBMAAP) [30,67,117–119], 3-(acrylamide)propyl 3-sulfopropyl dimethyl ammonium (SBAAAP) [121], dimethyl 4-vinylbenzyl 1-ammonio methanesulfonate (DVBAMS) [122], dimethyl 4-vinylbenzyl 3-ammonio propanesulfonate

(DVBAPS) [66–69,122], and dimethyl 4-vinylbenzyl 4-ammonio butanesulfonate (DVBABS) [122]. The carboxybetaine monomers involve 2-(methacryloyloxy)ethyl 1-carboxymethyl dimethyl ammonium (C₂BMA) [124–130,140,141], 2-(methacryloyloxy)ethyl 2-carboxylethyl dimethyl ammonium (C₃BMA) [22,108,110,131–135], 3-(methacrylamide)propyl 1-carboxymethyl dimethyl ammonium (C₂BMAA) [109], 3-(methacrylamide)propyl 2-carboxylethyl dimethyl ammonium (C₃BMAA) [109,136], 3-(acrylamide)propyl 2-carboxylethyl dimethyl ammonium (C₃BAA) [137,138] and (2-hydroxy-3-methacryloyloxy-propyl) dimethyl 2-ammonio acetate (CBOH) [139]. The other betaine types include 4-vinylbenzyl [122,142,143], 4-vinylpyridium [144] and norbornene derivatives [145], as well as zwitterionic PEI dendrimers [150–152].

3. Fabrication methods

Various polymers can be anchored on plane, curved surfaces and particles via “graft-from” and “graft-to” strategies [6,153–156]. Certainly, these methods can be applied to modify various surfaces by the zwitterionic polymers as shown in Fig. 2. The “grafted-from” methods always require a first anchoring of the initiator on solid surface and a subsequent polymerization of monomers in solution, and they usually provide high polymer grafting density and stability as advantages [157,158]. In “graft-to” approaches, a beforehand synthesized zwitterionic polymer is immobilized on the solid surface via physical interactions [51,76,159–164] or chemical bonding [61,165,166]. Moreover, surface grafting copolymerization [39,62,103,105,135] and post zwitterionization methods are developed to construct zwitterionic polymer grafted surfaces [72,144,167–169]. In addition, it is also possible to manufacture a zwitterionic membrane for ultrafiltration by blending the membrane-made polymer and the zwitterionic polymer during the casting process of the membrane [169,170].

3.1. “Graft-from”

The “grafted-from” method usually provides a uniform polymer layer on the solid surface with high stability. Surface-initiated (SI) polymerization involves atom transfer radical polymerization (ATRP) [46,69–71,87–99,140–142], reverse addition-fragmentation chain transfer (RAFT) polymerization [52,65,120], physisorbed free radical polymerization [115,131], ring-opening metathesis polymerization [145], and photo- [57,98,139], thermal- [72], plasma- [40,64] initiated radical polymerization etc.

3.1.1. ATRP

The SI-ATRP method can accurately control the molecular weight and graft density of the polymer by changing the monomer concentration and reaction time [41–45,80,87], it also can adjust chemical composition of copolymer chains by varying comonomer ratio in copolymerization [171–176]. A polymer layer of polyMPC or polySBMA were prepared on the surface of porous silica microspheres as stationary phases by conventional SI-ATRP [55,88]. Choi et al. also grafted silica

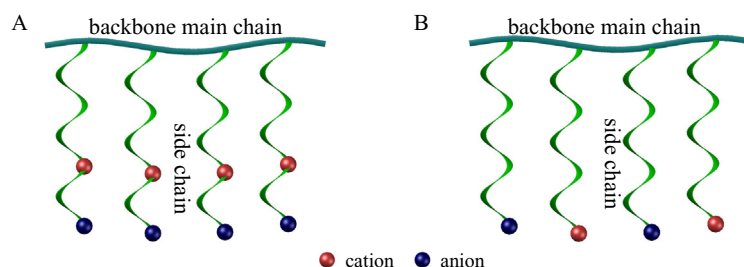


Fig. 1. Basic classification of (A) betaine-like type and (B) polyampholyte type of zwitterionic polymers.

Table 1
Structure, name, abbreviation and references of betaine-like monomers.

Monomer structure	Monomer name, abbr.	Polymer abbr.	References
	2-methacryloyloxyethyl phosphorylcholine, MPC	polyMPC	[39–73]
	2-(methacryloyloxy)ethyl 3-sulfopropyl dimethyl ammonium, SBMA	polySBMA	[69,73–116]
	2-(acrylamide)ethyl 3-sulfopropyl dimethyl ammonium, SBAE	polySBAE	[120]
	3-(methacrylamide)propyl 3-sulfopropyl dimethyl ammonium, SBMAAP	polySBMAAP	[30,67,117–119]
	3-(acrylamide)propyl 3-sulfopropyl dimethyl ammonium, SBAAP	polySBAAP	[121]
	dimethyl 4-vinylbenzyl 1-ammonio methanesulfonate, DVBAMS	polyDVBAMS	[122]
	dimethyl 4-vinylbenzyl 3-ammonio propanesulfonate, DVBAPS	polyDVBAPS	[66–69,122]
	dimethyl 4-vinylbenzyl 4-ammonio butanesulfonate, DVBABS	polyDVBABS	[122]
	2-(methacryloyloxy)ethyl 1-carboxymethyl dimethyl ammonium, C ₂ BMA	polyC ₂ BMA	[124–130,140,141]
	3-(methacrylamide)propyl 2-carboxylethyl dimethyl ammonium, C ₃ BMAA	polyC ₃ BMAA	[109,136]
	2-(methacryloyloxy)ethyl 2-carboxylethyl dimethyl ammonium, C ₃ BMA	polyC ₃ BMA	[22,108,110,131–134]
	3-(methacrylamide)propyl 1-carboxymethyl dimethyl ammonium, C ₂ BMAA	polyC ₂ BMAA	[109]
	3-(acrylamide)propyl 2-carboxylethyl dimethyl ammonium, C ₃ BAA	polyC ₃ BAA	[137,138]
	(2-hydroxy-3-methacryloyloxy)-propyl dimethyl 2-ammonio acetate, CBOH	polyCBOH	[139]
	3-[1-(4-vinylbenzyl)-1H-imidazol-3-ium-3-yl] propane sulfonate, VBIPS	polyVBIPS	[122,142,143]
	[4-(2-sulfethyl)][1-(4-vinylbenzyl)] pyridinium, SVBP	polySVBP	[122]
	4-vinyl-pyridinium-1-propane sulfonate, VPPS	polyVPPS	[144]
	5-norbornene-2-ylcarboxylethyl-1-imidazolium propane sulfonate, NCIPS	polyNCIPS	[145]

nanoparticles with crosslinked polyMPC shells by SI-ATRP using Mo (CO)₆ as catalyst from initiating trichloroacetyl group [56]. PolySBMA and polySBMAAP brushes were grafted from the gold surfaces with attached mercaptoundecyl bromoisobutyrate initiator by SI-ATRP, and the thickness of the obtained grafted polymer layers was approximately 20 nm [111,118]. Well-controlled polySBMA brushes were grafted on PES membrane from chloromethylated PES, in which the grafting yield of polySBMA was controlled by the reaction time [97]. Similarly, polySBMA brushes were also grafted on commercial polyamide

reverse osmosis (RO) membranes or cotton fabrics [81,90]. Liu and Ma's groups grafted polySBMA brushes on dopamine-based initiator modified surfaces of polyamide thin film composite (TFC) forward osmosis (FO) membranes, respectively [92,93,99]. In addition, a polySBMA brush was also grafted from the initiating surface of silicon wafer [96]. Ohno and coworkers constructed an initiating surface of silica or glass substrates with variable initiator density which resulted in polyC₂BMA brushes with gradually varied grafting density after a SI-ATRP reaction [140]. Russell and his coworkers reported a SI

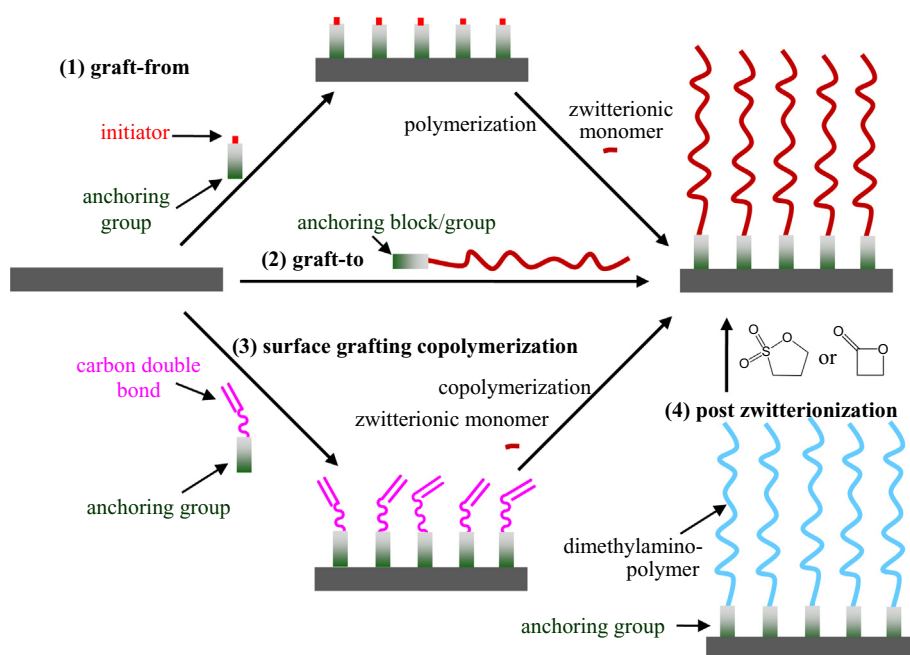


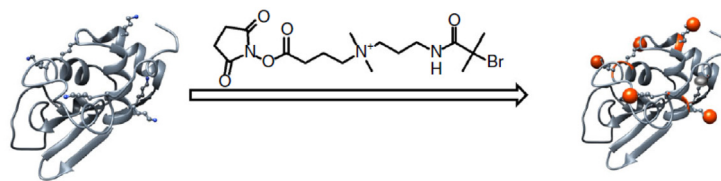
Fig. 2. Illustration for the fabrication strategies of the zwitterionic polymer grafted surfaces, including “graft-from”, “graft-to”, surface grafting copolymerization and post zwitterionization.

activator-regenerated-by-electron-transfer (ARGET) ATRP method to modify proteins with polyC₂BMA polymers [130]. First, as shown in Fig. 3, a positively charged ATRP initiator is chemically attached on a model protein lysozyme through a coupling reaction of an initiator-unit containing active maleimide compound with an amino group on the protein surface. Then, a SI-ARGET-ATRP reaction can be performed to grow the polyC₂BMA brush on the protein surface, and the obtained polyC₂BMA grafted protein has increased solubility in ammonium sulfate and completely prevents precipitation even in a saturated solution [130]. Moreover, the SI-ARGET-ATRP reaction was also used to graft the polySBMA brush on TFC membranes against biofouling [106,107].

Furthermore, a so-called sub-surface initiated (SSI) ATRP method was designed to graft a copolymer of zwitterionic SBMA and anionic

3-sulfopropyl methacrylate potassium to the surface of a hydrophilic resin substrate where a macromolecule initiator with an initiator group in each unit was pre-embedded [83,114]. The obtained copolymer brushes have better stability and durability than those polymer brushes prepared by common SI-ATRP, because the polymerization reaction in SSI-ATRP is performed from the surface to the sub-surface, forming a thick brush layer with tens micrometer [83]. The SSI-ATRP method was also used to fabricate polySBMA brushes on PVDF membrane [177]. Recently, polyMPC brushes were reported to graft the surface of silicon oxide by Cu(0)-mediated SI-ATRP [47] and polySBMA films were grafted from the surface of rough enzyme-absorbed electrode by electrochemically mediated ATRP to fabricate enzyme-based biosensors [89].

A ATRP initiator attachment



B ATRP

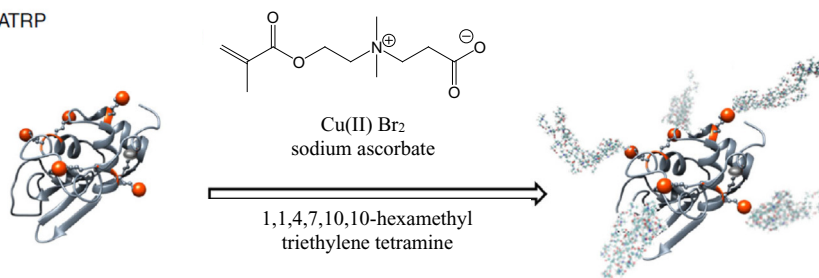


Fig. 3. Grafting polyC₂BMA onto a protein surface by SI-ATRP. (A) A positively charged ATRP initiator was first reacted with accessible amino groups on the protein's surface. (B) Next, ATRP was used to grow a polymer of zwitterionic CBMA (modified from Ref. [130]).

3.1.2. RAFT polymerization

Han and Tang used SI-RAFT polymerization of MPC to modify the surface of intraocular lens of silicone elastomer [65], Yu et al. prepared a zwitterionic polySBAAE modified stationary phase of porous silica particles for hydrophilic interaction liquid chromatography (HILIC) by SI-RAFT polymerization [120]. In addition, fluorescence-modified poly[MPC-co-(3-formyl-4-hydroxybenzyl methacrylate)] prepared by RAFT polymerization was used to functionalize mesoporous silica nanoparticles which showed good dispersity in water, excellent biocompatibility in organisms and remarkable light stability in bioimaging [53].

3.1.3. Physisorbed free radical polymerization

Zwitterionic polymer brushes were grafted onto the surface of PVDF membrane by a physisorbed free radical grafting polymerization in a heterogeneous reaction system of the PVDF membrane and aqueous monomer solution [115,131]. The initiator azobisisobutyronitrile (AIBN) can be mainly adsorbed on the surface of the PVDF membrane because of the hydrophobicity. In the mechanism, three possible reactions for the isobutyronitrile radicals will occur: (1) terminate each other between two radicals; (2) initiate a free radical polymerization in solution; and (3) abstract hydrogen from PVDF backbone. Then, the macro-radicals of PVDF backbone formed from hydrogen abstraction can react with the SBMA monomers in aqueous solution to form grafted polymer chains [115]. Such polymerization on PVDF membrane using AIBN as an initiator is chiefly on account of the hydrogen abstraction from the PVDF backbone, so that the polySBMA chains can directly grow from both the surface and the inside of the membrane [115].

3.1.4. Ring-opening metathesis polymerization

Zhang and Liu et al. designed a subsurface-initiated ring-opening metathesis polymerization to graft robust thick polyNCIPS brush onto the surface of polydimethylsiloxane rubber elastomer [145]. The grafting thickness of the polymer brush reached micron scale, and the as-grafted surface showed good lubricating, antimicrobial and antibiofouling properties [145].

3.1.5. Photo-, thermal- and Plasma- induced polymerization

A highly hydrophilic nanometer-scale modified surface was fabricated on a poly(ether ether ketone) substrate by photo-induced grafting polymerization [63]. The photo irradiation resulted in the generation of semibenzopinacol radical from benzophenone structure of the poly(ether ether ketone) and this radical acted as a photo-initiator during grafting polymerization of MPC [63]. Similar methods were used to graft crosslinked PE or PES membrane with different zwitterionic polymers [57,108,139]. Kyomoto et al. modified the surface of crosslinked PE by ultraviolet (UV) induced polymerization of MPC and the high grafting density of polyMPC on PE was essential for orthopedic bearing materials [57]. Weinman et al. grafted polyCBOH on PES membranes as switchable zwitterionic surfaces for biofouling control [139], and Guymon et al. fabricated polySBMA or polyC₃BMA functionalized glass substrate using Ronchi rule photo-mask for patterned samples or a cut glass slide for uniform samples [108]. In addition, polyC₃BMA grafted mesoporous silica films and polyDVBAPS grafted silicon wafers were also fabricated by the photo-induced polymerization [68,133].

Seo et al. developed a thermal-induced grafting method for polyMPC and polySBMA to achieve uniform, geometry-independent graft layer on curved crosslinked PE surfaces [72], which was expected to be useful in the development of complex, personalized and biocompatible artificial surfaces. Moreover, a plasma-initiated grafting polymerization also provided a simple and rapid manufacturing process for surface modification [40]. Liu et al. demonstrated that polyMPC could be grafted onto silicone contact intraocular lens by atmospheric plasma-initiated polymerization [64]. In addition, polySBMA was immobilized on the lumen surface of the PVDF hollow fiber membrane by an ozone activation strategy [86], and the grafting density and chain length of the polymer were systematically optimized by operating parameters.

3.2. "Graft-to"

The direct surface coating techniques in the "graft-to" strategy mainly include physical adsorption [59,76–78], chemical bonding [50] and biomimetic bonding [51,159]. The polymer grafting density in the "graft-to" methods is usually relatively low due to steric hindrance of large polymer chains.

3.2.1. Physical adsorption

Hydrophobic interaction and electrostatic interaction are usually employed to immobilize a zwitterionic copolymer onto a membrane surface [76–78]. First, hydrophobic interaction between the hydrophobic membrane surface and the hydrophobic block of zwitterionic-containing copolymer can easily facilitate the physical adsorption process [73,76]. For example, poly(SBMA-*b*-propylene oxide) can be self-assembled on the surface of PVDF microfiltration membranes [76] or on the 1-nonanethiol covered gold surface [74] due to the hydrophobic poly(propylene oxide) block. Second, as a representative of electrostatic interaction, poly[SBMA-*b*-(2-methacryloyloxy ethyl trimethylammonium chloride salt)] carrying positive charges was electrostatically adsorbed on the negatively charged surface of the membranes [77].

Moreover, low fouling bicontinuous membranes of PVDF with zwitterionic brushes were fabricated by in-situ modification method, so-called vapor-induced phase separation, from a casting solution of blending PVDF and triple copolymer poly(styrene-*co*-ethylene glycol methacrylate-*co*-SBMA), in which the hydrophobic interaction between polystyrene block and PVDF surface played a key role in the phase separation process [78]. Besides, a zwitterionic polymer network coating were prepared by spin-coating/drop-casting polymer solution onto 3-aminopropyl triethoxysilane modified silicon or silicon nitride surface with a subsequent drying [102].

3.2.2. Biomimetic bonding

Poly(MPC-*co*-dopamine-containing methacrylate) was anchored onto cell-membrane-mimetic interface via dopamine recognition [51]. Chen et al. used tannic acid and amphoteric polymer as inspiration to construct antifouling surface with resistance to protein adsorption, bacterial adhesion and platelet adhesion [159]. Firstly, tannic acid covered on the surface of PES film adsorbs Fe(III) ion to form a matrix layer; then partially quaternized PEI as zwitterionic polymer is anchored on the PES film through the reaction between the amino group of PEI and the oxidized catechol of tannic acid [159].

3.2.3. Chemical bonding

Click chemistry [119,125], amidation [50], epoxy ring opening reaction [113] and covalent Au-S bond formation [61,178] were utilized to build zwitterionic surfaces [50,61,125]. For example, Zhao et al. prepared poly[*N*-(*O*-nitrobenzyl acetate)-*N,N*-dimethyl ammonium ethyl methacrylate] to graft PES film by click chemistry of azide-alkyne cycloaddition [125]; Wu et al. reported a strategy to construct zwitterionic surfaces on the basis of interfacial thiol-ene click chemistry [119]. In addition, poly(MPC-*b*-lipoic methacrylate) could be anchored on the surface of gold nanorods by the formation of Au-S bond due to the presence of lipoic group [61], poly(SBMA-*co*-glycidyl methacrylate) was designed for the surface zwitterionization of versatile biomaterials including ceramic, metal and plastics, in which base-induced ring-opening reaction between epoxied and hydroxyl group was selected to immobilize the polymer onto the target surfaces [113].

3.3. Surface grafting copolymerization

The carbon double bond immobilized on a surface can be polymerized with the monomer in solution, resulting in a polymer grafted surface [39,62,135]. For instance, one-step distillation-precipitation polymerization has been developed as a typical surface grafting

copolymerization method [103,105], in which the unsaturated carbon double bond pre-fixed on the surface can polymerized with SBMA in aqueous solution. Similarly, the methacrylate group anchored on the gold electrode surface can reaction with carboxybetaine monomer in the solution under UV irradiation to produce a zwitterionic grafted gold surface [135].

3.4. Post zwitterionization

Tertiary amine groups can easily transform into quaternary ammonium, while they will transform into zwitterionic groups if reacting with 1,3-propanesultone or 1,3-propiolactone [165–169,179]. For example, polyDMAEMA brushes constructed on poly(ethylene terephthalate) sheets were partially quaternized to form zwitterionic poly(carboxybetaine) or poly(sulfobetaine) surfaces [167]. Similarly, the partial quaternization of polyDMAEMA with 1,3-propanesultone provided poly(DMAEMA-co-SBMA)-grafted mesoporous silica nanoparticles [168], and the quaternization reaction of poly(4-vinyl pyridine) brush on silicon with 1,3-propanesulfonate provided polyVPPS modified surface [144].

Yang et al. synthesized crosslinked poly[DMAEMA-co-(ethylene glycol dimethacrylate)] thin films on a RO membrane and then the polyDMAEMA block reacted with 1,3-propanesulfonate to form zwitterionic structures [166]. Jin et al. fabricated a PVDF membrane with blended poly[DMAEMA-co-(2-hydroxyethyl methacrylate)] and a subsequent sulfonation reaction transformed dimethylamino groups to zwitterionic SBMA units [169].

4. Hydrophilicity, salt affinity and other properties

The zwitterionic polymers can impart hydrophilicity to the grafted surfaces since the solvation of zwitterions on polymer chains will produce a high retention of solvent water and forms a deeply-hydrated layer around polymer chains through ion-dipole electrostatic interaction and electrostatic-induced hydrogen bonding [180]. Sum frequency generation (SFG) vibrational spectroscopy and quartz crystal microbalance with dissipation (QCM-D) techniques were typical microscopic methods for the behavior studies of the zwitterionic polymers [109–111,124]. The water contact angle measurement was commonly used to measure the surface hydrophilicity as reported in many literatures on the zwitterionic surfaces [38,64–66,98–101,117,118,131,144].

4.1. Microscopic analysis

SFG vibrational spectroscopy as a special interface method was used to probe the surface hydration properties of zwitterionic polymer brushes on silica prisms [109,110,124]. The zwitterionic polymer brushes are found to take a good ordering in water, and a strongly hydrogen-bonded water structure is demonstrated at polymer/water interfaces. However, in salt solution the results indicate a reduced ordering of the interfacial water due to the presence of counterions, and they also imply that the ions have different binding affinities to the zwitterionic polymers. For example, carboxybetaine binds Na^+ more strongly than K^+ , whereas sulfobetaine binds K^+ more strongly than Na^+ [109]. It is also revealed that all grafted polySBMA, polyC₂BMAA and polyC₃BMAA brushes on the silica prisms bind divalent cations more strongly than monovalent cations, and polyC₃BMAA binds Mg^{2+} and Ca^{2+} more strongly than polyC₂BMAA. Meanwhile, for the polyC₂BMAA and polyC₃BMAA brushes, lower pH leads to weaker ordering of the interfacial water due to the deionization of carboxyl groups, while the ordering of water at polySBMA/water interface is not affected by pH changes [109]. The pH dependent deionization of carboxyl groups in the poly(carboxybetaine) results in positively charged surface under acidic conditions but zwitterionically charged surface under basic conditions, thus the poly(carboxybetaine) grafted surfaces should have different hydrophilicity with pH changes [133].

The recent pH-dependent studies of the SFG vibrational spectroscopy in a pH range from 2 to 12 show that the absolute orientations of water molecules on zwitterionic surfaces grafted with the polyC₃BMA and polySBMA brushes are differently affected by pH change [110]. At the polyC₃BMA/water interface, water molecules have same orientations at pH values of 2 and 7, but have an opposite orientation at pH of 12. However, on the polySBMA/water interface, water molecules have the same orientations at pH of 7 and 12, but lose the ordering at pH of 2. This phenomenon can be explained by the different values of pK_a for the carboxybetaine and sulfobetaine groups. The orientation of water molecules is related to the surface charge and the water binding strength of positive/negative groups on the polymer surface [110]. The time dependent examination of salt effects on the hydration of the zwitterionic polymers indicates that the polyC₃BMA surface and the associated interfacial water exhibit a slow restructuring process after salt binding (likely due to the strong binding strength of polyC₃BMA with water), whereas the surface of polySBMA and the associated water have a fast change after salt binding [110]. The results of SFG vibrational spectroscopy provide important knowledge on the detailed water structures associated with the zwitterionic polymer brushes under different environments in addition to the strong hydration, which will be very helpful to comprehend their antifouling behaviors on molecular mechanism.

On the other hand, the conformational behavior of the zwitterionic polymer brushes was also investigated by using QCM-D technique [111]. For polySBMA brushes, it is found that the effectiveness of anions to weaken the inter/intra chain association and to enhance the hydration of grafted chains increases from kosmotrope to chaotrope in low ionic strength regime, but the ordering of anions is almost reversed at high ionic strengths. This dissipation change indicates that some heterogeneous structures are formed inside the brushes in the presence of chaotropic anions with the increase of ionic strength [111]. However, no obvious cation specificity can be observed in the conformational change of the brushes. In addition, the surface plasmon resonance (SPR) method was also attempted to measure polySBMA brushes, but no anion specificity was observed in the measurement because the change of resonance unit should be not sensitive to the coupled water molecules inside the brushes [111].

4.2. Hydrophilicity

The zwitterionic polymers are very hydrophilic due to their unique interactions with water, which can be measured by the obvious decreasing of water contact angle, static or dynamic, on the grafted surfaces compared with the virginal surfaces [66]. The water contact angle on polyMPC grafted silicone contact lens decreased to 50–60° from original 90° [64]. The static water contact angle on zwitterionic polyurethane (PU) surface modified with polySBMA brushes reduced to near 20° from pristine 90°, implying successful hydrophilization of PU surface [100].

Many superhydrophilic surfaces are obtained by grafting the zwitterionic polymers [117,118]. The water contact angle on acrylic resin discs grafted with thick, crosslinked polySBMAAP reached to a minimum value of 6.7°, close to a superhydrophilic surface, at 15% content of the zwitterionic monomer in the copolymerization [117]. The instantaneous water contact angle on polyMPC-modified intraocular lens surface was about 34°, then continually decreased to less than 5° (corresponding a superhydrophilic surface) after equilibrium for 6 min in dynamic measurement, while those data for the unmodified surface were around 113° [65]. Moreover, polyMPC-grafted glass, silicon, steel, polypropylene and polytetrafluoroethylene surfaces presented water contact angles of 7°, 12°, 19°, 18° and 17°, respectively [50]. The water contact angle of Ti alloy plate grafted by poly[SBMA-co-(methacryloyloxymethyl phosphonic acid)] decreased from original 59.6° to 12–15°; however water drop disappears immediately on such

Ti alloy plate grafted with poly[SBMA-*b*-(methacryloyloxymethyl diethyl phosphonate)], indicating a maximal superhydrophilicity [101].

The surface of porous polypropylene membrane with densely tethered polySBMA displayed a minimum water contact angle of 17.4°, much lower than that of 106° for untreated membrane [98]. The water contact angle on a polySBMA-grafted TFC FO membrane displayed a function of ATRP reaction time, the polySBMA-grafted membrane achieved a smallest water contact angle of about 8° (near to superhydrophilic wetting characteristic) as prolonging the polymerization time [99]. In other works, the water contact angle on polySBMA-grafted TFC membrane decreased from original 74° to 21° [92], and that on polyC₃BMA-grafted PVDF membranes gradually decreased from original approximately 91° till to superhydrophilic 0° along with the polymer grafting density in dynamic measurement [131].

4.3. Salt responsive mechanism and salt affinity

The solvation of anions and cations in the zwitterionic polymer attracts a large amount of water around the polymer chain, so the zwitterionic surface has a high hydrophilicity; meanwhile, these anions and cations electrostatically bind the counterions in salt solutions and affect the surface hydrophilicity. In pure water, as shown in Fig. 4A, the cations and anions on the polymer brush chains adsorb abundant of solvent water through ionic solvation and hydrogen bonding interaction, the ionic pairing of hydrated cations and anions of intra-monomer, intra-chain and inter-chain due to electrostatic attraction simultaneously makes the polymer brush take a relatively shrunk conformation. However, in salt solution the counterions penetrate into the polymer brush and are adsorbed onto the anions and cations (Fig. 4B); this destroys the previous electrostatic attraction of the charge pairs of intra-monomer, intra-chain and inter-chain. Then, the zwitterionic polymer brush postures more extended conformation, meanwhile the hydrodynamic radius of the polymer chains increases, companied by an obvious increase of viscosity [142]. The thickness of the zwitterionic polymer brush from AFM measurement obviously increases in NaCl aqueous solution as compared with in pure water [37,67]. That is to say, the zwitterionic polymer brush shrinks in water but stretches in salt solution, so they have greater solubility in brine than in pure water. Such salt-responsive behavior of the zwitterionic brush essentially reveals a strong salt affinity of the zwitterionic polymer, which is completely opposite to the polyelectrolyte [15,181], so that it is called as anti-polyelectrolyte behavior [143,182].

The salt-responsive behavior, i.e. salt affinity of the zwitterionic polymer brush makes itself occupy more solvent water in and around polymer chains through ionic solvation and hydrogen bond interaction of salt counterions with water, and brings the grafted surface with more superhydrophilicity at higher salt concentration [143]. As far as we know, almost all water in nature and biological systems has certain salinity, so the zwitterionic surfaces are factually used in aqueous saline solutions. The results of regular and phase-sensitive SFG vibrational

spectroscopy demonstrate that the zwitterionic polymer brush has a stronger affinity to divalent cations than monovalent cations [109], this should be ascribed to the stronger electrostatic attraction. It is believed that the affinity of the zwitterionic polymer brush to trivalent ions (such as Al³⁺, Fe³⁺ and PO₄³⁻) should be stronger than that of the bivalent and monovalent ions, thus these trivalent ions may have possibly unique effects on the properties of the zwitterionic surfaces.

Although the high hydrophilicity and strong salt affinity of the grafted surfaces are mainly determined by the cation and anion groups in the zwitterionic polymers, it is not difficult to understand that the monomers with different nonionic segments will affect the hydrophilicity and salt affinity of the grafted surfaces. For example, cellulose membranes grafted with different zwitterionic polymers showed different water contact angles, approximately 21° for polySBMA and polyMPC, but 34° and 48° for polyDVBAPS [69] and polyVBIPS [143], respectively. Such difference in the surface hydrophilicity could be explained by the more hydrophobicity of the benzyl groups in polyDVBAPS and polyVBIPS than the ester groups in polySBMA and polyMPC. Next, Quintana and Vancso compared zwitterionic polySBMAAP and polyDVBAPS grafted silicon wafers, the results demonstrated that the polySBMAAP grafted surface had better hydrophilicity and higher salt retention than the polyDVBAPS grafted surface [67], the reason could also be ascribed to the better hydrophilicity of the amide groups than the benzyl groups. Overall, betaine with cation and anion groups in a polymer unit mainly determines the hydrophilicity of the zwitterionic surface, while the nonionic segment gives a subsidiary contribution. In addition, zwitterionic polymer brushes were found to be stable in incubation with water or salt solutions, and they showed reversible wettability switching in response to salt concentration and counterion type [111,143].

4.4. Antifouling properties

The hydrophilicity and even superhydrophilicity of the zwitterionic polymers makes the grafted surface lack of hydrophobicity, so that severely weakens the interaction between the surface and those substrates majorly containing hydrophobic components. Therefore, the zwitterionic polymer grafted surfaces have shown strong antifouling properties [94,123,142,144,166]. In biological systems the salt affinity of the zwitterionic polymer brush in physiological saline environment shall behave stronger hydration and result in more effectively antifouling property than in pure water. Meanwhile, the zwitterionic surfaces also meet other two prerequisites for resisting protein adsorption: (i) they are electrically neutral surfaces because of equivalent cationic and anionic functional groups in the polymers, which avoid the electrostatic attraction to biomolecules [85]; and (ii) they contain hydrogen bond acceptors but cannot provide hydrogen bond donors, because most zwitterionic units have not hydrogen bond donors, or for polyacrylamide, have no chance to dedicate their hydrogen bond donors due to the hindrance of the surrounding zwitterionic groups. As a result,

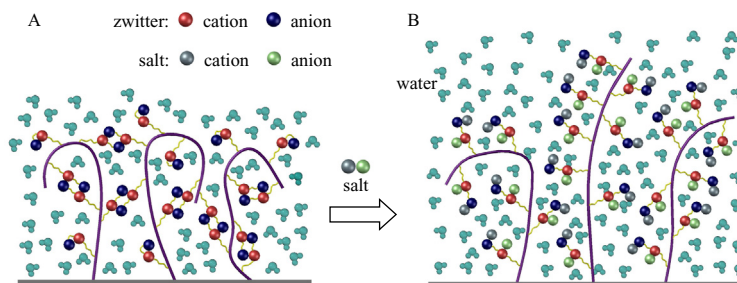


Fig. 4. Salt-response mechanism for anti-polyelectrolyte effect of zwitterionic polymer brushes.

the zwitterionic polymer modified surfaces have almost lost their capability to adhere/adsorb to various foulants such as biomolecules and their assemblies (protein, cell and blood component) [29,177,183]. Such ability of the zwitterionic surfaces to resist and repel the adsorption or adhesion of biological contaminants, i.e. so-named antibiofouling property, is very important for biological applications.

The zwitterionic surfaces can effectively prevent the adhesion of many biomolecules and display a low-contamination behavior, so that they can avoid the surface induced biological reactions [25,26]. For example, the amount of both adsorbed proteins and adhered cells on polyC₂BMA brush-covered substrate was easily controlled by the gradation of the surface density of zwitterionic brushes [140]. Similarly, polyC₃BMA was demonstrated to inhibit adsorption of non-specific proteins [22], poly(SBMA-*b*-propylene oxide) adsorbed on a hydrophobic substrate effectively prevented fibrinogen adsorption [74]. Because of the resistance to proteins, Jiang and Luo had to use EDC/NHS chemistry to immobilize enzyme protein on polyC₃BMA grafted gold surfaces [22] and electrode surfaces [135], respectively. Moreover, surface plasmon resonance (SPR) studies showed that zwitterionic polymer brushes effectively resisted protein adsorption in the presence of a more chaotropic anion, while a more chaotropic anion more effectively induced the protein desorption from zwitterionic brush surface [111].

4.5. Lubricating properties

The zwitterionic polymer grafted surfaces showed good lubrication performance due to the high water retention of hydrated polymer layers [37,41–46,75,79,143,161–164]. PolySBMA as an effective boundary lubricant on the surface of blended poly(vinyl alcohol) hydrogel reduced the friction coefficient by more than 80% compared with unblended hydrogel [79]. Typically, the tribology of grafted silicon wafers with the zwitterionic polymers was reported by Zheng and Yang et al. [68,143]. The friction coefficients of polyVBIPS and polyDVBAPS brushes can be effectively controlled by changing the concentration, cation and anion types of salts as shown in Fig. 5. The friction coefficients of the zwitterionic surfaces decrease monotonously to a very-low value when increasing the concentration of the NaCl solution

(Fig. 5A and D), which implies a better lubricating properties at higher salt concentration [68,143]. Then, the friction coefficients of polyVBIPS and polyDVBAPS brushes are closely related to the type of anions and cations in the solution. Both polyVBIPS and polyDVBAPS brushes reveal consistent order of anions of SO_4^{2-} , Cl^- , Br^- and NO_3^- (Fig. 5B and E) [68,143]. The strong electrostatic attraction of bivalent SO_4^{2-} to the positive charge on the polymer chain may push water molecules away and reduce the hydration of polymer chains, resulting lower surface hydration and higher surface friction. However, the Cl^- , Br^- and NO_3^- ions induce lower surface friction. In another case, the friction coefficient of polySBMA brush has been found to only increase slightly from 0.006 to 0.01 when changing anions from Cl^- , ClO_4^- , PF_6^- to bis (trifluoromethane sulfonimide) in the salt solution, which indicates a similar role for monovalent anions [75]. For the influence from cations, remarkably, the friction coefficient of polyVBIPS grafted surface in 1.0 mol/L NaCl solution is approximately 0.39 (Fig. 5C), which is larger than that of about 0.04, 0.05 and 0.09 in 1.0 mol/L KCl, MgCl_2 and CaCl_2 , respectively [143]. However, for polyDVBAPS grafted surface (Fig. 5F), the comparable friction coefficients, i.e. 0.015, 0.024, 0.01 and 0.019 respectively for Na^+ , K^+ , Mg^{2+} and Ca^{2+} , indicate a minor specificity of the polyDVBAPS brush to cations [68]. Although the bivalent Mg^{2+} and Ca^{2+} have stronger electrostatic attraction with negative ions on the zwitterionic polymers than the monovalent Na^+ and K^+ , all their comparable friction coefficients can possibly be ascribed to the strong hydration capability of Mg^{2+} and Ca^{2+} which may overcome the opposite action. It should be noticed that the ultralow friction surface of the zwitterionic polymer brush can also be obtained in water or dilute salt solution at higher temperature because of the better solubility of polyDVBAPS, thus temperature is considered as another parameter for tuning surface friction [68].

A recent work investigated the hydrophilicity and salt-responsiveness of several zwitterionic brushes, polyDVBAMS, polyDVBAPS, polyDVBABS, polySVBP and polyVBIPS, with different cationic moieties in their monomer units and different lengths of carbon spacers between cation and anion [122]. The effects of the subtle structural changes on the surface hydration and friction were explored both in water and in salt solutions of different concentrations and counterion

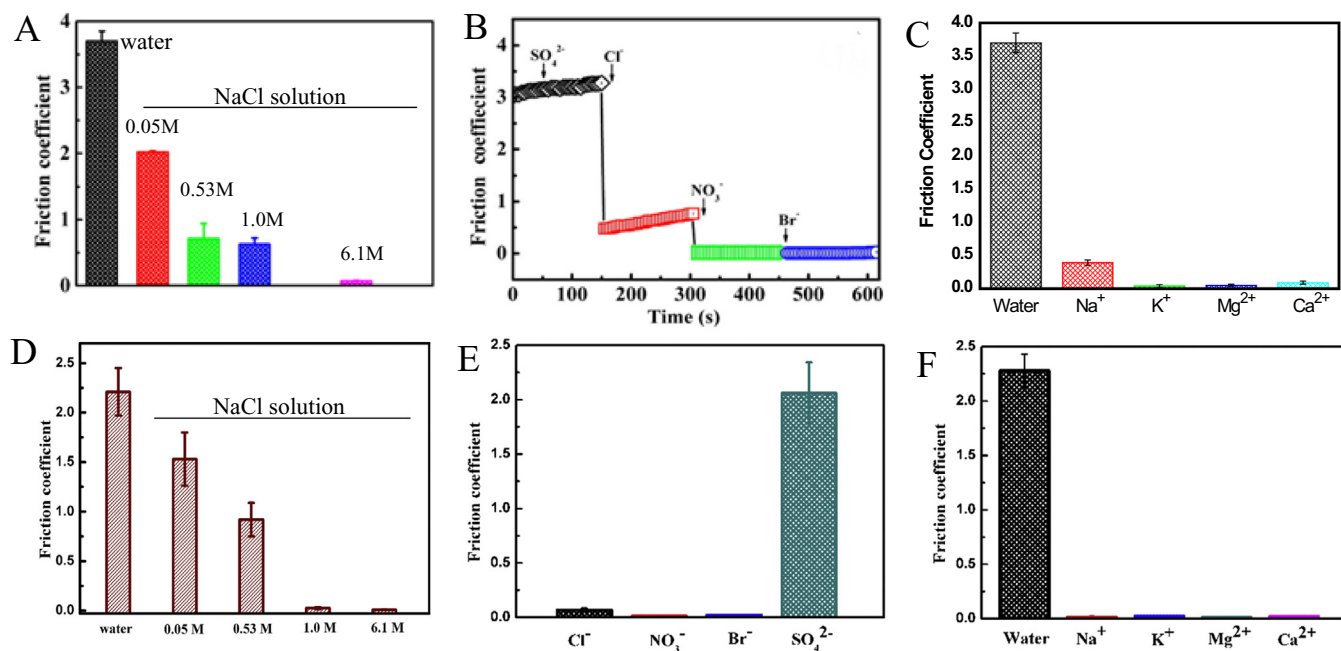


Fig. 5. The friction coefficient changes of polyVBIPS brush (A–C) and polyDVBAPS brush (D–F) in different concentration of NaCl solution (A, D), different anion solution (B, E) and cation solution (C, F) (modified from Ref. [68, 143]).

types. The collective results showed that the zwitterionic brushes with different cationic moieties and shorter carbon spacer lengths in salt solution induced higher surface friction and lower surface hydration than those in water, exhibiting strong salt responsive behaviors [122]. Moreover, by tuning the carbon spacer length, cationic moiety, salt concentration and type, the water contact angle on the grafted surfaces was changed from $\sim 60^\circ$ to $\sim 9^\circ$, while the interfacial friction can be changed from ultrahigh friction (friction coefficient ~ 4.5) to superior lubrication ($\sim 10^{-3}$) [122].

5. Applications

The surfaces covered by a monolayer of zwitterionic molecules were recently reported to have better antifouling property, cell uptake and biocompatibility than PEGylated surfaces, they also showed good performance in separating highly polar biomolecules [184–187]. As an alternative to the widely used PEG in biomedical field, the zwitterionic polymers have excellent antibiofouling property [25,26,74], so the zwitterionic surfaces not only can prevent the adsorption of biomolecules and their assemblies [25,26], but also dramatically decrease the adhesion of live organisms (especially microbes) [28,29,96]. Thereby, they are considered to be a class of passive antibacterial surface that reduce bacterial adhesion but cannot kill the bacteria [188,189]. Many reported works in biological related fields demonstrate that the surface modification with the zwitterionic polymers has enhanced the antibacterial properties of the materials [59,129,138,142]. For example, because the phosphorylcholine unit in polyMPC have a structure similar to that of phosphorylcholine lipid on the surface of living cells, the phosphorylcholine polymers show excellent hydrophilicity, biocompatibility and long-term stability, so they are believed to be increasingly used as mimetic cell membranes for creating bio-interfaces [48–50,190,191].

5.1. Grafted membranes for separation

5.1.1. Plasma separation

The hydrophilicity, antifouling and antibacterial properties of the zwitterionic surfaces have shown significance in membrane separation of biological systems particularly for feasible, low-cost plasma ultrafiltration [69,170,192]. The zwitterionic membranes have proved to be biocompatible, hemocompatible and blood-inert surfaces and display great potential in the separation and biomedical fields [97,116]. For example, compared to the original membranes, polyMPC-bearing copolymer modified PES membrane, chitosan membrane and cellulose membranes showed a 93% water flux recovery after three cycles of fouling with bovine serum albumin (BSA) in ultrafiltration [51], a 60% decrease of protein (BSA or fibrinogen) adsorption and a 96% suppression of platelet adhesion [48], and a prevention of adhesion, deformation and aggregation of platelets after contacting with platelet-rich plasma [54], respectively. The filtration membrane of the crosslinked polyMPC-containing copolymer via an electro-spinning process exhibited an obvious decrease in platelet adhesion and protein adsorption, and this membrane removed 80% of the protein and almost 98% of the platelet during the sequential filtration of platelet-rich plasma solutions [192].

Zwitterionic polySBMA grafted PES membrane or PVDF membrane showed lower BSA and fibrinogen adsorption, lower platelet adhesion but prolonged clotting time [97], or displayed enhanced fouling resistance after incubation with fibrinogen, bacteria and whole blood [78]. In permeability tests, after incubation in a platelet-poor-plasma solution the polySBMA-grafted PVDF membrane showed a flux recovery ratio of 91% which outperformed 66% for the commercial PVDF membrane [78]. Moreover, polySBMA immobilized on interior surface of PVDF hollow fiber membrane exhibited superior fouling resistance during the filtration of BSA solution [86], especially the irreversible fouling-induced

resistance was prominently reduced from 53% to 15% due to the outstanding hydration capacity of the polymer [86].

The protein adsorption tests and chemical force microscopy studies of polySBMA grafted PVDF ultrafiltration membranes indicated that a large brush layer thickness (>100 nm) was necessary to impart effective fouling resistance [94]. However, polySBMA modified membranes exhibited only a slight increase of water flux recovery during dynamic fouling tests with BSA. Despite the thick polySBMA brush layer, this insufficient water flux recovery indicated that internal fouling occurred in the membrane pores. To prevent the internal fouling, polySBMA brushes were grafted onto both the outer and internal pore surfaces of the membrane, but the dynamic fouling tests showed only a minute improvement in the water flux recovery again. Therefore, the final reason was ascribed to the small pore size of the membrane which imposed a fundamental limitation on the thickness of the brush layer inside the pores [94].

5.1.2. Water separation

Zwitterionic polymer grafted membranes for water separation comprise RO [81,165,166], FO [99,106,112] ultrafiltration membranes [51,139,193] and common filtration membranes [73,77,169]. The grafting of polySBMA brushes on commercial polyamide RO membranes enhanced the surface hydrophilicity, smoothness and denseness [81], they not only prevented static bacterial adhesion, but also effectively resisted the production of biofilm and biofouling. The grafted membranes with enough polymer thickness presented no permeability decline and little biofilm coverage [81]. The polySBMA brushes on polyamide FO membranes also showed significantly lower water flux decline than pristine membranes based on their excellent fouling resistance to the mixed organic foulants (BSA, alginate, and natural organic matter) [92]. A copolymer of zwitterionic SBMA and cationic 2-methacryloyloxy ethyl trimethylammonium chloride salt modified poly(vinyl chloride-co-acrylonitrile-co-sodium 4-styrenesulfonate) membrane performed anti-organic fouling and antibiofouling properties [77]. In addition, Wu et al. reported that zwitterionic PEI dendrimer endowed PES membrane with an electro-neutral separation layer and enhanced antifouling property [150], Zhao et al. reported that the zwitterionic PEI dendrimer modified PES surface with in-situ generated Ag nanoparticles displayed excellent resistance to protein adsorption and bacterial adhesion [151]. The polyCBOH grafted PES membranes performed better antifouling properties than other common antifouling surfaces in the bacteria deposition studies, and they only accumulated half the biovolume as unmodified membranes in the biofilm studies [139].

The membranes for industrial oil-containing wastewater separation always experience fouling problems after long-term use because of their weak hydrophilicity, which seriously impedes the development of membrane technologies for practical usage. To overcome this problem, a polySBMA brush was grafted onto a commercial TFC FO membrane for advanced treatment of oil/water emulsions [99]. The best polySBMA modified membrane exhibited superhydrophilicity and ultralow oil-adhesion. Dynamic fouling experiments using a feed solution of 10,000 ppm oil/water emulsion demonstrated a superior antifouling ability of the membrane, as displayed by lower water flux decline than the control membrane [99]. Furthermore, the multicycle FO fouling tests showed that the irreversible membrane fouling of the polySBMA modified membrane during long term use could be significantly alleviated through salt cleaning. The salt cleaning mechanism, as shown in Fig. 6, is attributed to the salt-responsive property of the polySBMA brushes, which will provide a driving force for the release of oil foulants from the membrane surface [99].

Poly(SBMA-co-trifluoroethyl methacrylate) and poly(MPC-co-trifluoroethyl methacrylate) were respectively used as selective layers to coat TFC PVDF membranes for the treatment of municipal and industrial wastewater streams [73]. These two membranes exhibited high rejections to organics and excellent resistance to fouling by organic

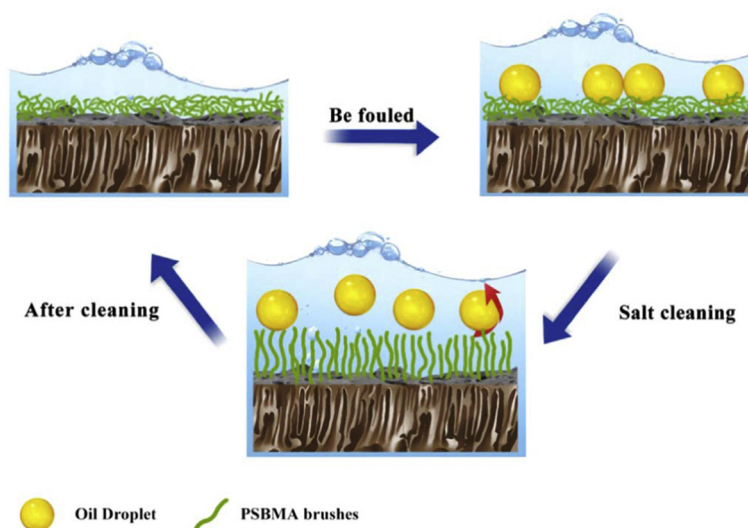


Fig. 6. The fouling process of polymer brushes grafted on forward osmosis membrane by oil droplet contaminant, the salt cleaning mechanism based on the salt-responsive property of zwitterionic polymer brushes (from reference [99]).

foulants in municipal wastewater, they also showed complete resistance to fouling by dyes and colored substances in industrial textile wastewater; meanwhile, salt was allowed to permeate through the membranes, providing an effluent with high salt recovery [73]. Moreover, the zwitterionic membranes were chemically stable and exhibited no change in performance upon exposure to acidic, basic and chlorinated cleaning solutions. Therefore, the excellent fouling resistance, the chemical resistance and the ability to effectively remove wastewater pollutants make these membranes very promising for wastewater treatment applications [73]. In addition, polyMPC modified PES ultrafiltration film and polySBMA-grafted PVDF membrane were used to separate water from oil-in-water emulsions with high water permeation flux and good antifouling property [51,169].

5.1.3. Salt separation

The salt affinity of the zwitterionic polymers provides a high salt retention within the polymer chains [67]. The polyC₃BMA layer grafted on PVDF membrane showed improved inorganic salts separation performance, hydrophilicity and antifouling property [131], the ultrafiltration experiments suggested that the modified membrane possessed stable initial sodium chloride rejection ratio and reversible flux behavior [131]. In addition, poly[MPC-co-(2-aminoethyl methacrylate)] was incorporated into polyamide selective layer of the TFC membrane to remove Se and As concurrently [60]. Comparing with the pristine TFC membrane, the zwitterionic copolymer modified TFC membrane presented improved pure water permeability and effective rejection for pollutants due to their reduced pore sizes and high hydrophilicity. It was found that the newly developed membrane containing 50 wt% of the copolymer exhibited the highest pure water permeability and nearly whole rejection to SeO_3^{2-} , SeO_4^{2-} and HAsO_4^{2-} , even in the simultaneous removal of Se and As ions [60].

5.2. Grafted stationary phases for HILIC column separation

In recent years, inorganic anions, cations, organic acid/bases and biomolecules have been rapidly separated by monolithic columns with dynamic coating, covalent bonding and ion exchange, where hydrophilic interaction liquid chromatography (HILIC) plays a key role as an effective strategy to separate polar chemical substrates [194–201]. In the zwitterionic polymer brushes hydrophilic-hydrophobic character, dipole-dipole interaction of anions and cations, as well as strong salt affinity to counterions in solution shall endow the grafted surfaces with

selective adsorption properties [55,84,202]. In 2002, Irgum et al. reported a tentacle-type zwitterionic stationary phase of porous silica particles with grafted polySBMA layer [84], the evaluation on the chromatographic columns packed with these particles indicated that this zwitterionic stationary phase could simultaneously and independently separate inorganic anions and cations when using aqueous solutions of perchloric acid or perchlorate salts as eluents [84]. Such stationary phase material was also capable of separating two acidic and three basic proteins in a single run using gradient salt elution at constant pH. In 2006, Irgum and Jiang's group reported polyMPC and polySBMA grafted stationary phases of porous silica particles, respectively [55,88]. The separation experiments of several peptides in the HILIC separation mode demonstrated that small peptides could be separated via the hydrophilic interaction and the ionic interaction between the stationary phase and the analytes, in which the organic solvent composition, pH and salt concentration of the eluent showed strong effect on the retention time [55].

Highly selective and efficient capture of glycoproteins and glycopeptides from complex biological samples is greatly significant for detecting and identifying disease biomarkers in biological systems. Zhang et al. reported that $\text{Fe}_3\text{O}_4/\text{SiO}_2$ nanoparticles were grafted with thick polySBMA shells and several functional groups were also attached on the polymer chains to improve the binding capacity and detection sensitivity for glycopeptides [103]. The resulting zwitterionic nanocomposites were demonstrated to have an outstanding ability for glycopeptide enrichment with extremely high detection sensitivity (0.1 fmol), large binding capacity (100 mg g⁻¹), high enrichment recovery (above 73.6%) and rapid magnetic separation. Furthermore, in the analysis of real complicated biological samples, 905 unique N-glycosylation sites from 458 N-glycosylated proteins were reliably identified in three replicate analyses of a 65 µg protein sample extracted from a mouse liver [103]. Besides, a similar work of polySBMA modified Fe_3O_4 nanoparticles also demonstrated high selectivity for glycopeptide enrichment [105].

Saccharides (such as fructooligosaccharide, trisaccharide isomers and ginsenosides) as challenging targets in HILIC mode were well separated on the zwitterionic stationary phase of polySBAAE modified porous silica particles [120]. This stationary phase material displayed good selectivity for the enrichment of glycopeptides in HILIC mode. As shown in Fig. 7A, glycopeptide signals without enrichment were severely suppressed by non-glycopeptides in electrospray ionization mass spectrometry (ESI-MS) analysis and only two glycopeptide signals

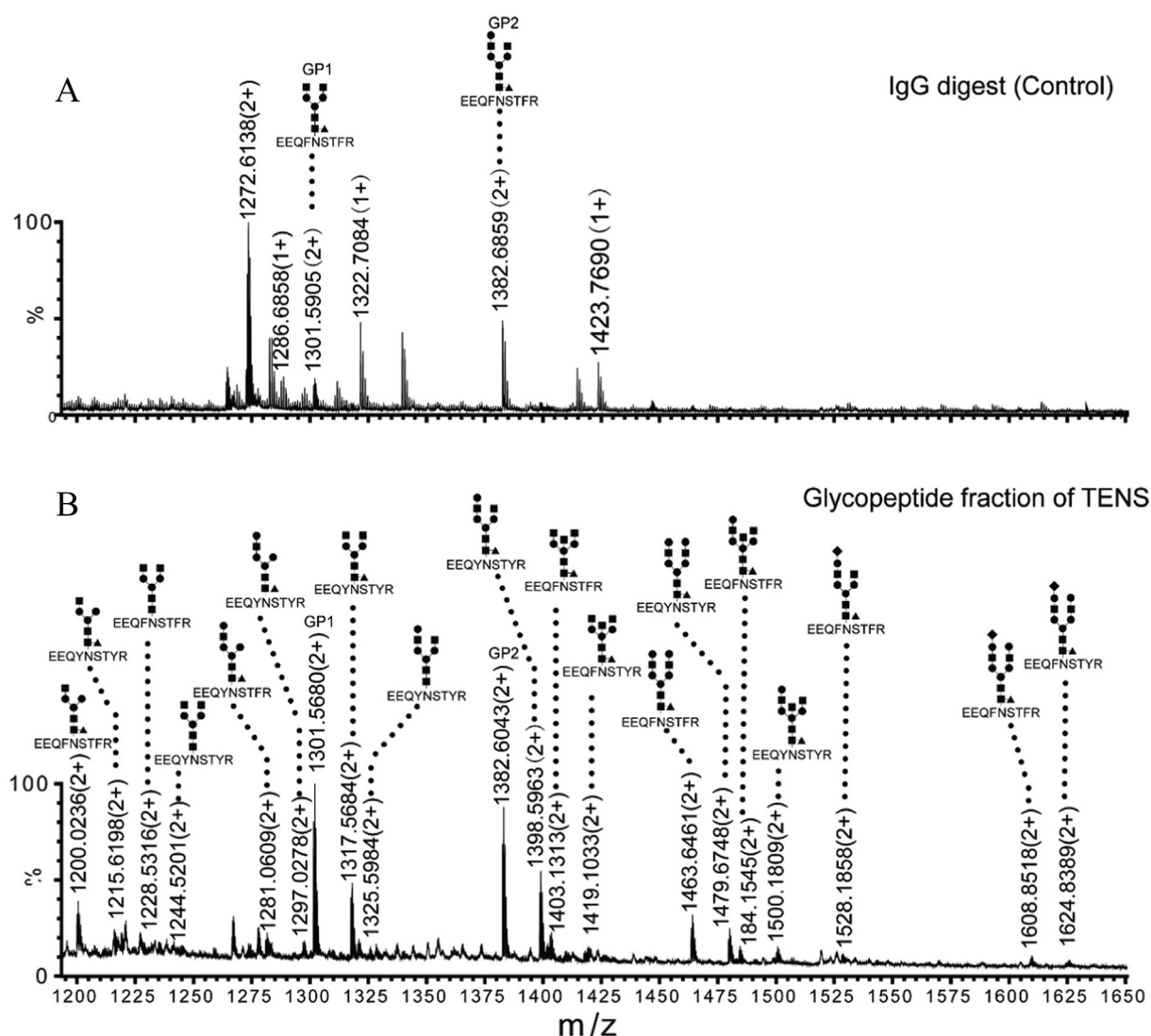


Fig. 7. Electrospray ionization mass spectrometry (ESI-MS) analysis of the human IgG digest before (A) and after the enrichment on zwitterionic porous silica stationary phase (B) (from Ref. [120]).

were detected. However, after enrichment with this stationary phase material, as shown in Fig. 7B, most of the non-glycopeptides were eliminated and the signals of the glycopeptides were remarkably improved, resulting in a total of twenty glycopeptides being detected [120].

Zwitterionic polySBMA-grafted micron-level beads of poly[(3-chloro-2-hydroxypropyl methacrylate)-co-(ethylene dimethacrylate)] were utilized as stationary phase for the separation of polar analytes in HILIC mode [95]. It was proved that the stationary phase carrying zwitterionic brushes with the longer chain length had better chromatographic separation result than others; the chain length of the zwitterionic brushes (i.e. molecular weight) was more effective with respect to its surface concentration (i.e. grafting density) for better separation of nucleosides, nucleotides, acrylic compounds and organic acids [95].

5.3. Grafted solid surfaces

5.3.1. Organic surfaces

The zwitterionic homopolymers and copolymers were widely applied to functionalize various surfaces to produce antibiofouling and antibacterial properties [82,87,91,100]. The copolymer brushes containing zwitterionic polySBMA and anionic poly(3-sulfopropyl methacrylate potassium) on hydrophilic resin substrate were reported to have better

stability and long-term synergistic antifouling effect in seawater medium through soaking experiment and ocean field assay [83], but the excellent antifouling performances of poly(3-sulfopropyl methacrylate potassium) brushes were hindered by salt responsiveness.

The polySBMA brushes grafted on cotton fabrics could enhance the hydrophilicity and hygienicity of textiles for high-quality life because of the nonfouling property and long-term resistance to biofilms [90]. The polyMPC-grafted silicone contact lens could decrease adsorbed proteins (human serum albumin and fibrinogen) by about 25–30% compared with pristine one [64,65].

5.3.2. Inorganic surfaces

Chang et al. reported that the zwitterionic PEI dendrimers on hydroxyapatite discs and human tooth showed an inhibition up to 70% of bacteria adhesion [152]. The discs of denture base acrylic resin were coated with a thick copolymer of sulfobetaine methacrylamide and *N,N'*-(4,7,10-trioxo-1,13-tridecadiamine), and the obtained zwitterionic discs were found to have superhydrophilicity and significantly reduced adhesion of microbes [117,123]. Bailey et al. demonstrated that polySBMA-grafted silicon wafer had better antifouling property to undiluted fetal bovine serum than the un-grafted surface [91], it also could significantly reduce the adhesion of bacteria [96]. The polyVBIPS

brushes on a gold substrate exhibited a reversible salt-induced switching property between biomolecule-adhesive state and biomolecule repellent state in complex media for multiple cycles, showing a switchable ability to resist both protein adsorption and bacterial attachment [142].

The antibiofouling and biocompatible properties of the zwitterionic polymers were utilized to repel cell adhesion and guide cell growth [108]. The investigation of polySBMA or polyC₃BMA grafted glass substrates with a uniform layer or a Ronchi rule pattern displayed that fibroblasts almost exclusively migrated and grew on uncoated bands, while only little or no cells presented on the zwitterionic bands, especially for polyC₃BMA-coated surfaces [108]. Moreover, Astrocytes and Schwann cells showed similarly low levels of cell adhesion and morphology changes when they were cultured on zwitterionic surfaces, while Schwann cells and inner ear spiral ganglion neuron neurites aligned well to the zwitterionic patterns [108]. Such approach should be significant for neural prosthetic implants with nearly eliminated fibrosis encapsulation.

5.3.3. Implanted material surfaces

Protein adsorption of the material surface always brings surface induced biological reactions, resulting in many adverse effects in the application of implanted materials. The zwitterionic surfaces can almost avoid the negative surface induced reactions due to high the hydrophilicity and antibiofouling properties, which is critical for implanting materials [58,85,89,101]. The polyMPC-modified alginate microspheres were demonstrated to have improved biocompatibility based on the reduction of the surface mediated fibrotic reaction [58]. Kwon et al. grafted PU surface with polySBMA-containing copolymer to prevent the thrombus formation or infection on the surface of biomaterial after transplantation [100]. The fluorescent images (Fig. 8A) indicated that the zwitterionic PU surface displayed stronger resistance to cell adhesion than the bare PU, and the result of counting adherent cells showed 91% reduction in fibroblast adhesion on the modified PU surface (Fig. 8B) [100].

PolyC₂BMA-modified Ti alloy plate exhibited strong suppression of protein adsorption and cell adhesion [128], and induced approximately twice the amount of calcium (Ca²⁺) deposition compared with the unmodified plate. The triblock copolymer brushes consisting of two zwitterionic polySBMA blocks and one poly(glycidyl methacrylate) block were grafted on the surfaces of Ti and stainless steel as implanting materials to improve their biofouling resistance [85]. The results showed that the copolymer with a polymerization number of 50 units in each

block (molecular weight ~15.2 kDa) behaved the best nonfouling property as assessed by plasma proteins, blood cells, fibroblast cells and various bacteria. This was attributed to the optimized grafting degree (molecular weight) and chain organization of the copolymer, because lower polymerization number (25) or higher values (75 or 100) might result in low surface coverage or formation of aggregates, respectively [85]. The metal surfaces grafted with the optimized copolymer showed efficient resistance to biofouling by a number of biofoulants including leukocyte concentrates, platelet concentrates, cells from whole blood, *Escherichia coli*, *Staphylococcus epidermidis*, and HT1080 fibroblast cells [85].

5.4. Nanomedicines

Surface properties of materials (such as charge and hydrophilicity) often determine the adsorption of proteins and other biomolecules, and the surface properties become the most important factors for the kinetics of drug delivery in the body [155,203]. The physiological properties of polyMPC in biomedical applications are comparable to natural cell membranes and can avoid many problems such as inflammatory infections. The poly(MPC-*b*-lipoic methacrylate) anchored on the surface of gold nanorods exhibited low dark-cytotoxicity to cells and long retention time in blood circulation [61]. Importantly, the high biocompatibility of the zwitterionic polymers makes drug delivery vehicles more “stealth” in blood circulation than PEG-based systems [204–206].

5.4.1. Zwitterionic therapeutic proteins

Modification of therapeutic proteins is an attractive approach to conflict many challenging diseases, but the obtained proteins are often unstable or have lowered bioavailability. Although the conjugation with PEG is often used to increase the stability of proteins, this operation always has a detrimental effect on bioactivity. However, the therapeutic protein conjugation with zwitterionic poly(carboxybetaine) shows improved stability in a manner similar to PEGylation, and the protein conjugates retain or even improve binding affinity to substrates [127]. Jiang et al. demonstrated that the polyC₂BMA conjugated model proteins offered better stability without sacrificing the binding affinity and bioactivity of the proteins [127]. The mechanism of how PEG and polyC₂BMA affect binding affinity is illustrated in Fig. 9. A general mechanism of molecule binding has two species with higher solubility for each other than their environment (Fig. 9A). For proteins, hydrophobic-hydrophobic interactions are significant in the interaction with other substrates. Attaching polymers (such as PEG) to the protein

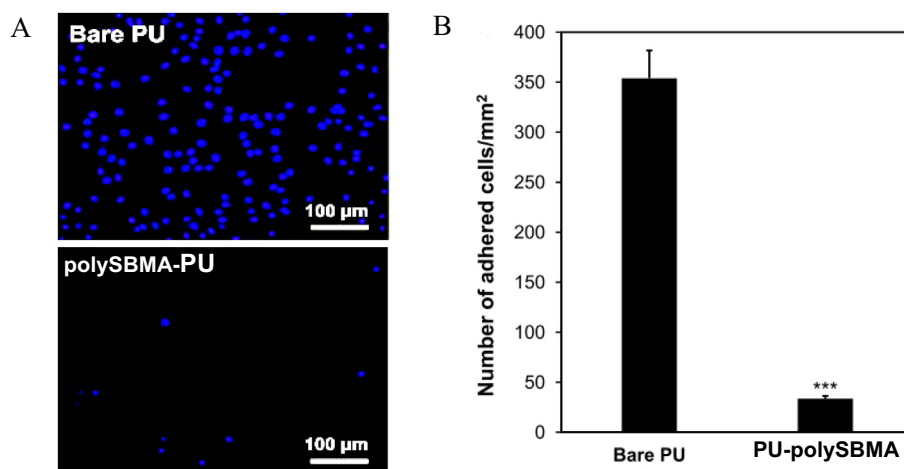


Fig. 8. (A) Fluorescence microscopy images of fibroblasts adhering on bare polyurethane (PU) and polySBMA modified PU substrates. (B) Counting number of cell nuclei on bare and modified PU substrates (from Ref. [100]).

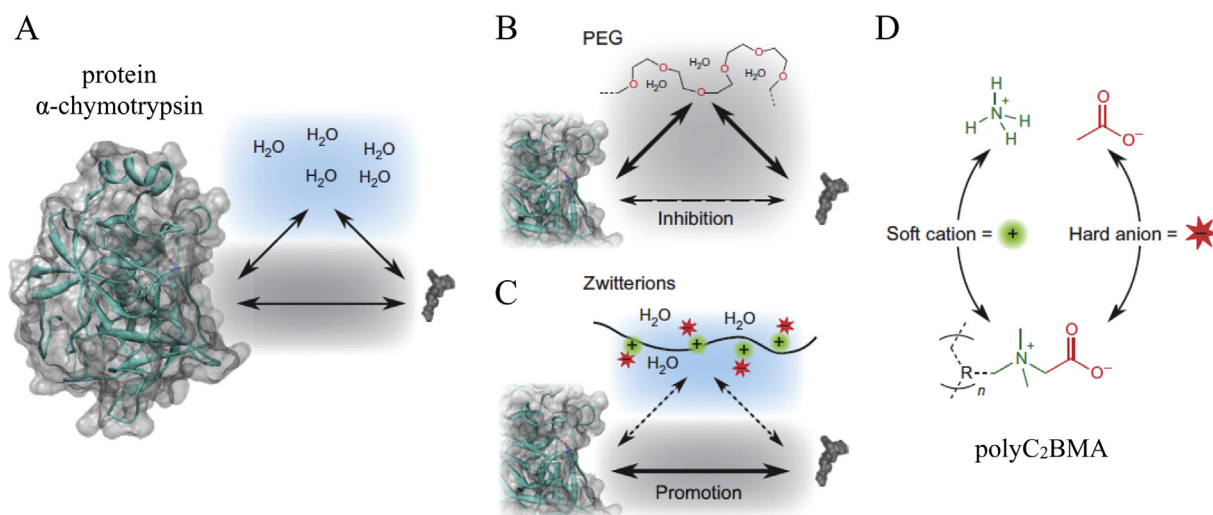


Fig. 9. Mechanism of how PEG and polyC₂BMA affect binding affinity. (A) Relationship between enzyme and substrate without polymer. (B) PEG impedes affinity by reducing enzyme-substrate hydrophobic-hydrophobic interactions. (C) Super-hydrophilic polyC₂BMA has a strong effect on the structure of water, creating a local environment that increases enzyme-substrate hydrophobic-hydrophobic interactions. (D) The structural relationship between polyC₂BMA and ammonium, acetate, both of which contain protein stabilizing ions found in the Hofmeister series (from ref. [127]).

surface imposes steric hindrances and creates competitive interactions with substrates and binding sites (Fig. 9B), thereby reducing binding affinity. In addition to the steric hindrances, the hydrophilic (not superhydrophilic) PEG chains can interfere with the substrate binding via reducing the hydrophobic-hydrophobic driving forces for substrate-site interaction. In contrast, with the superhydrophilic poly(carboxybetaine) conjugates (Fig. 9C), the binding affinity is either unaltered or possibly improved. Owing to the superhydrophilicity of the zwitterionic polymer chains, water is drawn away from hydrophobic regions of the proteins, which shifts the equilibrium to allow the interaction of substrates and binding sites. Thereby, the zwitterionic polymer chains can increase the substrate's affinity for the binding pocket. Moreover, the molecular structure (Fig. 9D) can explain the mechanism by which polyC₂BMA enhances binding affinity, because the betaine monomer is seen to correspond to the most stabilizing and active monovalent ions in the series of ammonium and acetate. This zwitterionic-protein conjugation opens a new avenue for the development of protein therapeutics by avoiding the need for a compromise between stability and affinity [127].

Moreover, the core-shell nanoparticles of crosslinked polyMPC grafted model protein showed a long circulation lifetime because the zwitterionic polymer brushes had strong resistance to nonspecific adsorption due to the phosphorylcholine structure similar to lipids in the outer membrane of living cells [39].

5.4.2. Grafted solid nanoparticles

Zwitterionic polymer-modified nanoparticles have a good "stealth" effect, so that they can effectively avoid agglomeration, reduce tumor clearance and increase tumor accumulation [205,206]. Jiang et al. reported that polyC₃BAA grafted silica nanoparticles had long-term stability in both negative and positive protein solutions [137]. PolyMPC grafted silica nanoparticles with encapsulated fluorophores were easily up-taken into the cytoplasm of HeLa cells by endocytosis, and the observed cells still maintain their normal morphology, demonstrating good biocompatibility and low cytotoxicity of the zwitterionic silica nanoparticles [71].

5.5. Miscellaneous cases

For the wound dressing application, Chang and his coworkers compared two wound-contacting membranes of expanded poly

(tetrafluoroethylene) grafted with zwitterionic polySBMA and hydrophilic PEGMA [40]. The results showed that the polySBMA-grafted membrane presented high hydration capability and high nonbioadhesive character in contact with protein solution, human blood, tissue cells and bacterial medium. The zwitterionic membrane dressing provided a moist environment, essential for "deep" skin wound healing [40]. In another work, zwitterionic hydrogels containing polySBAAP and silver nanoparticles exhibited relatively high water content, excellent non-sticky property, excellent resistance to BSA adsorption, negligible cytotoxicity and germicidal properties [121]. The results in the feasibility evaluation showed that the polySBAAP/silver hydrogels did not adhere to the newly formed tissue and were readily removed from the wounds, showing a great significance for the treatment of infected chronic wounds [121]. In addition, the zwitterionic polymers should have more potential in the molecular design of antibioadhesive membranes for skin tissue regeneration.

Moreover, crosslinked polyMPC was coated on silica nanoparticles for a purpose of crude oil attraction [56]. The core flooding experiment showed that the fluid of polyMPC grafted silica nanoparticles increased the oil recovery rate by 5 vol% under reduced injection pressure, because the zwitterionic nanoparticles could change the wettability of oil droplets on the surface of rock pores [56]. In addition, after mechanical scratching damage, the crosslinked zwitterionic polymer coating on silicon or silicon nitride surfaces showed fast self-healing capability when immersed in water [102].

6. Summary and outlook

In summary, a variety of zwitterionic polymers can be grafted on the surfaces of many substrates, including metal gold, Ti alloy and stainless steel, inorganic silicon wafer, silica and glass, and lots of organic polymer membranes or solid surfaces. The "grafting-from" and "grafting-to" strategies can graft well-defined zwitterionic polymer layers onto the surfaces via physical or chemical processes. The typical living radical polymerization methods (SI-ATRP and RAFT polymerization) can produce uniform zwitterionic polymer brushes because they can accurately control the polymer molecular weight and grafting density on the surfaces, even SSI ATRP can form thick brush layers to tens micrometer. The surface grafting copolymerization and post zwitterionization provide diverse selections for the zwitterionic functionalization of surfaces. Nevertheless, the fabrication techniques

for the zwitterionic polymer grafted surfaces will still attract much attention in the next years to improve the properties and lower the cost. The zwitterionic polymer grafted surfaces with superhydrophilicity, salt affinity, and antibiofouling, antibacterial properties can prevent from the surface-induced biological reactions on the material surfaces, which broaden the application area of plasma ultrafiltration, substance separation, implanting materials, drug delivery and so on. Especially, the zwitterionic brush grafted membranes are expected to be more effective in the selective separation.

The zwitterionic grafted surfaces should be considered as excellent materials for many potential applications. For example, they were used as support materials to fabricate bioelectronics and biosensors due to their biocompatibility and versatility. Enzyme-based electrochemical biosensors with polySBMA coating are used as promising implantable biosensors with long-time stability [89], polyC₃BMA modified surfaces as matrix for biomolecules can develop ultrasensitive and low fouling biosensors [134–136]. Furthermore, the zwitterionic surfaces are expected to resolve underwater corrosion of marine facilities with expanding scale, because they are speculated to repel the adhesion of many organisms, including fungus, algae, protozoa, mollusks, etc. [83] However, the large-scale industrial practices of the zwitterionic grafted surfaces have not yet been realized by viable techniques, so more attention should be paid on this emerging topic in the future.

Moreover, the localized electric field between anions and cations in the zwitterionic polymer layers may interact with external electromagnetic field, which may be related with the density of the anion-cation pairs, the length of carbon spacers between anion and cation, and the media between the polymer chains. The zwitterionic polymer coatings are believed to have unexpected properties in strong electromagnetic field, and possibly the covered surfaces may be shielded from strong electromagnetic waves or light.

Declaration of competing interest

None.

Acknowledgements

This work was supported by the National Nature Science Foundation of China (NNSFC 21975139), and the Natural Science Foundation of Shandong Province (ZR2017MB042), China.

References

- [1] Tuson HH, Weibel DB. Bacteria-surface interactions. *Soft Matter* 2013;9:4368–80.
- [2] Banerjee I, Pangule RC, Kane RS. Antifouling coatings: recent developments in the design of surfaces that prevent fouling by proteins, bacteria, and marine organisms. *Adv Mater* 2011;23:690–718.
- [3] Ding X, Yang C, Lim TP, Li YH, Engler AC, Hedrick JL, et al. Antibacterial and antifouling catheter coatings using surface grafted PEG- b -cationic polycarbonate diblock copolymers. *Biomaterials* 2012;33:6593–603.
- [4] Yan JJ, Bockstaller MR, Matyjaszewski K. Brush-modified materials: Control of molecular architecture, assembly behavior, properties and applications. *Prog Polym Sci* 2020;100:101180.
- [5] Li DX, Li QR, Hao XW, Zhang YJ, Zhang ZP, Li CF. Assembled core-shell nanostructures of gold nanoparticles with biocompatible polymers toward biology. *Curr Top Med Chem* 2014;14:595–616.
- [6] Li DX, He Q, Li JB. Smart core/shell nanocomposites: Intelligent polymers modified gold nanoparticles. *Adv Colloid Interf Sci* 2009;149:28–38.
- [7] Ducker R, Garcia A, Zhang J, Tao C, Zauscher S. Polymeric and biomacromolecular brush nanostructures: progress in synthesis, patterning and characterization. *Soft Matter* 2009;4:1774–86.
- [8] Hiromi K, Takuya K, Tomohiro K, Shintaro I, Makoto N, Kohji O. Anti-biofouling properties of an amphoteric polymer brush constructed on a glass substrate. *Colloids Surf B: Biointerfaces* 2011;88:455–62.
- [9] Li DX, Li CF, Wang AH, He Q, Li JB. Hierarchical gold/copolymer nanostructures as hydrophobic nanotanks for drug encapsulation. *J Mater Chem* 2010;20:7782–7.
- [10] Tang DC, Zeng ZJ, Xia YQ, Chen BH, Gao S, Cao MW, et al. The effects of thermoresponsive microgel density on cell adhesion, proliferation, and detachment. *J Appl Polym Sci* 2020;137:48773.
- [11] Chi H, Wang MY, Xiao YT, Wang FK, Joshy KS. Self-assembly and applications of amphiphilic hybrid POSS copolymers. *Molecules* 2018;23:2481.
- [12] Ostuni E, Chapman RG, Holmlin RE, Takayama S, Whitesides GM. A survey of structure—property relationships of surfaces that resist the adsorption of protein. *Langmuir* 2001;17:5605–20.
- [13] Li L, Chen S, Jiang S. Protein interactions with oligo(ethylene glycol) (OEG) self-assembled monolayers: OEG stability, surface packing density and protein adsorption. *J Biomater Sci Polym Ed* 2007;18:1415–27.
- [14] Harris JM, Martin NE, Modi M. Pegylation: a novel process for modifying pharmacokinetics. *Clin Pharmacokinet* 2001;40:539–51.
- [15] Mary P, Bendejacq DD, Labeau MP, Dupuis P. Reconciling low- and high-salt solution behavior of sulfobetaine polyzwitterions. *J Phys Chem B* 2007;111:7767–77.
- [16] Sin MC, Chen SH, Chang Y. Hemocompatibility of zwitterionic interfaces and membranes. *Polym J* 2014;46:436–43.
- [17] Izquierdo-Barba I, Colilla M, Vallet-Regi M. Zwitterionic ceramics for biomedical applications. *Acta Biomater* 2016;40:201–11.
- [18] Bernards MT, Cheng G, Zhang Z, Chen S, Jiang S. Nonfouling polymer brushes via surface-initiated, two-component atom transfer radical polymerization. *Macromolecules* 2008;41:4216–9.
- [19] Schlenoff JB. Zwitterion: coating surfaces with zwitterionic functionality to reduce nonspecific adsorption. *Langmuir* 2014;30:9625–36.
- [20] Zhang WL, Li GJ, Lin YL, Wang LY, Wu SQ. Preparation and characterization of protein-resistant hydrogels for soft contact lens applications via radical copolymerization involving a zwitterionic sulfobetaine comonomer. *J Biomater Sci Polym Ed* 2017;28:1935–49.
- [21] Vaisocherová H, Zhang Z, Yang W, Cao Z, Cheng G, Taylor AD, et al. Functionalizable surface platform with reduced nonspecific protein adsorption from full blood plasma – material selection and protein immobilization optimization. *Biosens Bioelectron* 2009;24:1924–30.
- [22] Zhang Z, Chen S, Jiang S. Dual-functional biomimetic materials: nonfouling poly(carboxybetaine) with active functional groups for protein immobilization. *Biomacromolecules* 2006;7:3311–5.
- [23] Estephan ZG, Schlenoff PS, Schlenoff JB. Zwitterion As an Alternative to PEGylation. *Langmuir* 2011;27:6794–800.
- [24] Ladd J, Zhang Z, Chen S, Hower JC, Jiang S. Zwitterionic polymers exhibiting high resistance to nonspecific protein adsorption from human serum and plasma. *Biomacromolecules* 2008;9:1357–61.
- [25] Hadidi M, Zydney AL. Fouling behavior of zwitterionic membranes: impact of electrostatic and hydrophobic interactions. *J Membr Sci* 2014;452:97–103.
- [26] Yang W, Bai T, Carr LR, Keefe AJ, Xu J, Xue H, et al. The effect of lightly crosslinked poly(carboxybetaine) hydrogel coating on the performance of sensors in whole blood. *Biomaterials* 2012;33:7945–51.
- [27] Robert L, Tirrell DA. Designing materials for biology and medicine. *Nature* 2004;428:487–92.
- [28] Chen S, Jiang S. A new avenue to nonfouling materials. *Adv Mater* 2008;20:335–8.
- [29] Mi L, Jiang SY. Integrated antimicrobial and nonfouling zwitterionic polymers. *Angew Chem Int Ed* 2014;53:1746–54.
- [30] Cho WK, Kong B, Park HJ, Kim J, Chegal W, Choi JS, et al. Long-term stability of cell micropatterns on poly((3-(methacryloylamino)propyl)-dimethyl(3-sulfopropyl) ammonium hydroxide)-patterned silicon oxide surfaces. *Biomaterials* 2010;31:9565–74.
- [31] Jiang S, Cao Z. Ultralow-fouling, functionalizable, and hydrolyzable zwitterionic materials and their derivatives for biological applications. *Adv Mater* 2010;22:920–32.
- [32] Chen S, Li L, Chao Z, Jie Z. Surface hydration: principles and applications toward low-fouling/nonfouling biomaterials. *Polymer* 2010;51:5283–93.
- [33] Zhang Z, Vaisocherová H, Cheng G, Yang W, Xue H, Jiang S. Nonfouling behavior of polycarboxybetaine-grafted surfaces: structural and environmental effects. *Biomacromolecules* 2008;9:2686–92.
- [34] Zhang Z, Chao T, Chen S, Jiang S. Superlow fouling sulfobetaine and carboxybetaine polymers on glass slides. *Langmuir* 2006;22:10072–7.
- [35] Xia Y, Adibnia V, Huang R, Murschel F, Fèvre J, Xie G, et al. Biomimetic bottlebrush polymer coatings for fabrication of ultralow fouling surfaces. *Angew Chem Int Ed* 2019;58:1308–14.
- [36] Lee W-F, Tsai C-C. Synthesis and solubility of the poly(sulfobetaine)s and the corresponding cationic polymers: 1. Synthesis and characterization of sulfobetaines and the corresponding cationic monomers by nuclear magnetic resonance spectra. *Polymer* 1994;35:2210–7.
- [37] Kobayashi M, Terayama Y, Kikuchi M, Takahara A. Chain dimensions and surface characterization of superhydrophilic polymer brushes with zwitterion side groups. *Soft Matter* 2013;9:5138–48.
- [38] Kobayashi M, Terayama Y, Yamaguchi H, Terada M, Murakami D, Ishihara K, et al. Wettability and antifouling behavior on the surfaces of superhydrophilic polymer brushes. *Langmuir* 2012;28:7212–22.
- [39] Wang N, Jin X, Zhu XY. Construction of biomimetic long-circulation delivery platform encapsulated by zwitterionic polymers for enhanced penetration of blood-brain barrier. *RSC Adv* 2017;7:20766–78.
- [40] Jhong JF, Venaunt A, Hou CC, Chen SH, Wei TC, Zheng J, et al. Surface zwitterionization of expanded poly(tetrafluoroethylene) membranes via atmospheric plasma-induced polymerization for enhanced skin wound healing. *ACS Appl Mater Interfaces* 2013;5:6732–42.
- [41] Chen M, Briscoe WH, Armes SP, Klein J. Lubrication at physiological pressures by polyzwitterionic brushes. *Science* 2009;323:1698–701.
- [42] Tairy O, Kampf N, Driver MJ, Armes SP, Klein J. Dense, highly hydrated polymer brushes via modified atom-transfer-radical-polymerization: structure, surface interactions, and frictional dissipation. *Macromolecules* 2015;48:140–51.

- [43] Kobayashi M, Terayama Y, Hosaka N, Kaido M, Suzuki A, Yamada N, et al. Friction behavior of high-density poly(2-methacryloyloxyethyl phosphorylcholine) brush in aqueous media. *Soft Matter* 2007;3:740–6.
- [44] Zhang Z, Morse AJ, Armes SP, Lewis AL, Geoghegan M, Leggett GJ. Effect of brush thickness and solvent composition on the friction force response of Poly(2-(methacryloyloxy)ethylphosphorylcholine) brushes. *Langmuir* 2011;27:2514–21.
- [45] Zhang Z, Morse AJ, Armes SP, Lewis AL, Geoghegan M, Leggett GJ. Nanoscale contact mechanics of biocompatible polyzwitterionic brushes. *Langmuir* 2013;29:10684–92.
- [46] Yu Y, Vancso GJ, Sd Beer. Substantially enhanced stability against degrafting of zwitterionic PMPC brushes by utilizing PGMA-linked initiators. *Eur Polym J* 2017;89:221–9.
- [47] Yan WQ, Fantin M, Ramakrishna S, Spencer ND, Matyjaszewski K, Benetti EM. Growing polymer brushes from a variety of substrates under ambient conditions by Cu-0-mediated surface-initiated ATRP. *ACS Appl Mater Interfaces* 2019;11:27470–7.
- [48] Gong M, Dang Y, Wang YB, Yang S, Winnik FM, Gong YK. Cell membrane mimetic films immobilized by synergistic grafting and crosslinking. *Soft Matter* 2013;9:4501–8.
- [49] Iwasaki Y, Ishihara K. Cell membrane-inspired phospholipid polymers for developing medical devices with excellent biointerfaces. *Sci Technol Adv Mater* 2012;13:064101.
- [50] Xing C-M, Meng F-N, Quan M, Ding K, Dang Y, Gong Y-K. Quantitative fabrication, performance optimization and comparison of PEG and zwitterionic polymer antifouling coatings. *Acta Biomater* 2017;59:129–38.
- [51] Xue Q, Cao H, Meng F, Quan M, Gong Y. Cell membrane mimetic coating immobilized by mussel-inspired adhesion on commercial ultrafiltration membrane to enhance antifouling performance. *J Membr Sci* 2017;528:1–11.
- [52] Liu Y, Mao L, Yang S, Liu M, Huang H, Wen Y, et al. Synthesis and biological imaging of fluorescent polymeric nanoparticles with AIE feature via the combination of RAFT polymerization and post-polymerization modification. *Dyes Pigments* 2018;158:79–87.
- [53] Chen J, Liu M, Huang L, Huang H, Wan Q, Tian J, et al. Preparation of zwitterionic polymers functionalized fluorescent mesoporous silica nanoparticles through photoinduced surface initiated RAFT polymerization in the presence of oxygen. *J Taiwan Inst Chem Eng* 2018;91:570–7.
- [54] Ishihara K, Shinozuka T, Hanazaki Y, Iwasaki Y, Nakabayashi N. Improvement of blood compatibility on cellulose hemodialysis membrane: IV. Phospholipid polymer bonded to the membrane surface. *J Biomater Sci Polym Ed* 1999;10:271–82.
- [55] Jiang W, Fischer G, Girmay Y, Irgum K. Zwitterionic stationary phase with covalently bonded phosphorylcholine type polymer grafts and its applicability to separation of peptides in the hydrophilic interaction liquid chromatography mode. *J Chromatogr A* 2006;1127:82–91.
- [56] Choi SK, Son HA, Kim HT, Kim JW. Nanofluid enhanced oil recovery using hydrophobically associative zwitterionic polymer-coated silica nanoparticles. *Energy Fuel* 2017;31:7777–82.
- [57] Kyomoto M, Moro T, Konno T, Takadama H, Yamawaki N, Kawaguchi H, et al. Enhanced wear resistance of modified cross-linked polyethylene by grafting with poly(2-methacryloyloxyethyl phosphorylcholine). *J Biomed Mater Res A* 2007;82:10–7.
- [58] Yesilyurt V, Veisheh O, Doloff JC, Li J, Bose S, Xie X, et al. A facile and versatile method to endow biomaterial devices with zwitterionic surface coatings. *Adv Healthc Mater* 2017;6:1601091.
- [59] Sae-Ung P, Kolewe KW, Bai Y, Rice EW, Schiffman JD, Emrick T, et al. Antifouling stripes prepared from clickable zwitterionic copolymers. *Langmuir* 2017;33:7028–35.
- [60] He Y, Liu J, Han G, Chung T-S. Novel thin-film composite nanofiltration membranes consisting of a zwitterionic co-polymer for selenium and arsenic removal. *J Membr Sci* 2018;555:299–306.
- [61] Jiang HY, Chen D, Guo DB, Wang N, Su Y, Jin X, et al. Zwitterionic gold nanorods: low toxicity and high photothermal efficacy for cancer therapy. *Biomater Sci* 2017;5:686–97.
- [62] Kyomoto M, Moro T, Iwasaki Y, Miyaji F, Kawaguchi H, Takatori Y, et al. Superlubricious surface mimicking articular cartilage by grafting poly(2-methacryloyloxyethyl phosphorylcholine) on orthopaedic metal bearings. *J Biomed Mater Res A* 2009;91A:730–41.
- [63] Kyomoto M, Ishihara K. Self-initiated surface graft polymerization of 2-methacryloyloxyethyl phosphorylcholine on poly(ether ether ketone) by photoirradiation. *ACS Appl Mater Interfaces* 2009;1:537–42.
- [64] Chen JS, Ting YS, Tsou HM, Liu TY. Highly hydrophilic and antibiofouling surface of zwitterionic polymer immobilized on polydimethylsiloxane by initiator-free atmospheric plasma-induced polymerization. *Surf Coat Technol* 2018;344:621–5.
- [65] Han YM, Xu X, Tang JM, Shen CH, Lin QK, Chen H. Bottom-up fabrication of zwitterionic polymer brushes on intraocular lens for improved biocompatibility. *Int J Nanomedicine* 2017;12:127–35.
- [66] Wu JH, Zhang D, Zhang LX, Wu BY, Xiao SW, Chen F, et al. Long-term stability and salt-responsive behavior of polyzwitterionic brushes with cross-linked structure. *Prog Org Coat* 2019;134:153–61.
- [67] Guo SF, Quintana R, Cirelli M, Toa ZSD, Vasantha VA, Kooij ES, et al. Brush swelling and attachment strength of barnacle adhesion protein on zwitterionic polymer films as a function of macromolecular structure. *Langmuir* 2019;35:8085–94.
- [68] Fu YH, Zhang L, Huang L, Xiao SW, Chen F, Fan P, et al. Salt- and thermo-responsive polyzwitterionic brush prepared via surface-initiated photoiniferter-mediated polymerization. *Appl Surf Sci* 2018;450:130–7.
- [69] Liu PS, Chen Q, Wu SS, Shen J, Lin SC. Surface modification of cellulose membranes with zwitterionic polymers for resistance to protein adsorption and platelet adhesion. *J Membr Sci* 2010;350:387–94.
- [70] Chen M, Briscoe WH, Armes SP, Cohen H, Klein J. Ployzwitterionic brushes: extreme lubrication by design. *Eur Polym J* 2011;47:51–23.
- [71] Mao LC, Liu XH, Liu MY, Huang L, Xu DZ, Jiang RM, et al. Surface grafting of zwitterionic polymers onto dye doped AIE-active luminescent silica nanoparticles through surface-initiated ATRP for biological imaging applications. *Appl Surf Sci* 2017;419:188–96.
- [72] Lim C-M, Hur J, Jang H, Seo J-H. Developing a thermal grafting process for zwitterionic polymers on cross-linked polyethylene with geometry-independent grafting thickness. *Acta Biomater* 2019;85:180–91.
- [73] Bengani-Lutz P, Zaf RD, Culfaz-Emecen PZ, Asatekin A. Extremely fouling resistant zwitterionic copolymer membranes with ~ 1nm pore size for treating municipal, oily and textile wastewater streams. *J Membr Sci* 2017;543:184–94.
- [74] Chang Y, Chen S, Zhang Z, Jiang S. Highly protein-resistant coatings from well-defined diblock copolymers containing sulfobetaines. *Langmuir* 2006;22:2222–6.
- [75] Wei QB, Cai MR, Zhou F, Liu WM. Dramatically tuning friction using responsive polyelectrolyte brushes. *Macromolecules* 2013;46:9368–79.
- [76] Venault A, Chang Y, Yang HS, Lin PY, Shih YJ, Higuchi A. Surface self-assembled zwitterionization of poly(vinylidene fluoride) microfiltration membranes via hydrophobic-driven coating for improved blood compatibility. *J Membr Sci* 2014;454:253–63.
- [77] Wang S-Y, Fang L-F, Cheng L, Jeon S, Kato N, Matsuyama H. Improved antifouling properties of membranes by simple introduction of zwitterionic copolymers via electrostatic adsorption. *J Membr Sci* 2018;564:672–81.
- [78] Dizon GV, Venault A. Direct in-situ modification of PVDF membranes with a zwitterionic copolymer to form bi-continuous and fouling resistant membranes. *J Membr Sci* 2018;550:45–58.
- [79] Osaheni AO, Finkelstein EB, Mather PT, Blum MM. Synthesis and characterization of a zwitterionic hydrogel blend with low coefficient of friction. *Acta Biomater* 2016;46:245–55.
- [80] Chen L, Honma Y, Mizutani T, Liaw DJ, Gong JP, Osada Y. Effects of polyelectrolyte complexation on the UCST of zwitterionic polymer. *Polymer* 2000;41:141–7.
- [81] Yang Z, Saeki D, Matsuyama H. Zwitterionic polymer modification of polyamide reverse-osmosis membranes via surface amination and atom transfer radical polymerization for anti-biofouling. *J Membr Sci* 2018;550:332–9.
- [82] Ye Z, Zhang P, Zhang J, Deng L, Zhang J, Lin C, et al. Novel dual-functional coating with underwater self-healing and anti-protein-fouling properties by combining two kinds of microcapsules and a zwitterionic copolymer. *Prog Org Coat* 2019;127:211–21.
- [83] Liu H, Ma Z, Yang W, Pei X, Zhou F. Facile preparation of structured zwitterionic polymer substrate via sub-surface initiated atom transfer radical polymerization and its synergistic marine antifouling investigation. *Eur Polym J* 2019;112:146–52.
- [84] Jiang W, Irgum K. Tentacle-type zwitterionic stationary phase prepared by surface-initiated graft polymerization of 3-[N,N-Dimethyl-N-(Methacryloyloxyethyl)-ammonium] propanesulfonate through peroxide groups tethered on porous silica. *Anal Chem* 2002;74:4682–7.
- [85] Chou YN, Venault A, Cho CH, Sin MC, Yeh LC, Jhong JF, et al. Epoxylated zwitterionic triblock copolymers grafted onto metallic surfaces for general biofouling mitigation. *Langmuir* 2017;33:9822–35.
- [86] Tang YP, Cai T, Loh D, O'Brien GS, Chung TS. Construction of antifouling lumen surface on a poly(vinylidene fluoride) hollow fiber membrane via a zwitterionic graft copolymerization strategy. *Sep Purif Technol* 2017;176:294–305.
- [87] Chang Y, Liao SC, Higuchi A, Ruaan RC, Chu CW, Chen WY. A Highly stable nonbiofouling surface with well-packed grafted zwitterionic polysulfobetaine for plasma protein repulsion. *Langmuir* 2008;24:5453–8.
- [88] Hemström P, Szumski M, Irgum K. Atom-transfer radical graft polymerization initiated directly from silica applied to functionalization of stationary phases for high-performance liquid chromatography in the hydrophilic interaction chromatography mode. *Anal Chem* 2006;78:7098–103.
- [89] Hu YC, Liang B, Fang L, Ma GL, Yang G, Zhu Q, et al. Antifouling zwitterionic coating via electrochemically mediated atom transfer radical polymerization on enzyme-based glucose sensors for long-time stability in 37 degrees C serum. *Langmuir* 2016;32:11763–70.
- [90] Wu J, Zhang C, Xu S, Pang X, Cai G, Wang J. Preparation of zwitterionic polymer-functionalized cotton fabrics and the performance of anti-biofouling and long-term biofilm resistance. *Colloid Interface Sci Commun* 2018;24:98–104.
- [91] Limpoco FT, Bailey RC. Real-time monitoring of surface-initiated atom transfer radical polymerization using silicon photonic microring resonators: implications for combinatorial screening of polymer brush growth conditions. *J Am Chem Soc* 2011;133:14864–7.
- [92] Liu CH, Lee J, Ma J, Elimelech M. Antifouling thin-film composite membranes by controlled architecture of zwitterionic polymer brush layer. *Environ Sci Technol* 2017;51:2161–9.
- [93] Liu CH, Lee J, Small C, Ma J, Elimelech M. Comparison of organic fouling resistance of thin-film composite membranes modified by hydrophilic silica nanoparticles and zwitterionic polymer brushes. *J Membr Sci* 2017;544:135–42.
- [94] Davenport DM, Lee J, Elimelech M. Efficacy of antifouling modification of ultrafiltration membranes by grafting zwitterionic polymer brushes. *Sep Purif Technol* 2017;189:389–98.
- [95] Kip C, Tuncel A. HILIC stationary phase based on monodisperse-porous polymethacrylate beads functionalized with zwitterionic molecular brushes. *Chromatographia* 2017;80:565–75.

- [96] Tang ZW, Ma CY, Wu HX, Tan L, Xiao JY, Zhuo RX, et al. Antiadhesive zwitterionic poly-(sulphobetaine methacrylate) brush coating functionalized with triclosan for high-efficiency antibacterial performance. *Prog Org Coat* 2016;97:277–87.
- [97] Yue W-W, Li H-J, Xiang T, Qin H, Sun S-D, Zhao C-S. Grafting of zwitterion from polysulfone membrane via surface-initiated ATRP with enhanced antifouling property and biocompatibility. *J Membr Sci* 2013;446:79–91.
- [98] Zhao YH, Wee KH, Bai R. Highly hydrophilic and low-protein-fouling polypropylene membrane prepared by surface modification with sulfobetaine-based zwitterionic polymer through a combined surface polymerization method. *J Membr Sci* 2010;362:326–33.
- [99] Zhang XY, Tian JY, Gao SS, Shi WX, Zhang ZB, Cui FY, et al. Surface functionalization of TFC FO membranes with zwitterionic polymers: Improvement of antifouling and salt-responsive cleaning properties. *J Membr Sci* 2017;544:368–77.
- [100] Kwon HJ, Lee Y, Phuon LT, Seon GM, Kim E, Park JC, et al. Zwitterionic sulfobetaine polymer-immobilized surface by simple tyrosinase-mediated grafting for enhanced antifouling property. *Acta Biomater* 2017;61:169–79.
- [101] Huang T, Liu HW, Liu PM, Liu PS, Li L, Shen J. Zwitterionic copolymers bearing phosphonate or phosphonic motifs as novel metal-anchorable anti-fouling coatings. *J Mater Chem B* 2017;5:5380–9.
- [102] Wang ZH, van Andel E, Pujari SP, Feng HH, Dijkman JA, Smulders MMJ, et al. Water-repairable zwitterionic polymer coatings for anti-biofouling surfaces. *J Mater Chem B* 2017;5:6728–33.
- [103] Chen Y, Xiong Z, Zhang L, Zhao J, Zhang Q, Peng L, et al. Facile synthesis of zwitterionic polymer-coated core-shell magnetic nanoparticles for highly specific capture of N-linked glycopeptides. *Nanoscale* 2015;7:3100–8.
- [104] Kitano H, Suzuki H, Matsuura K, Ohno K. Molecular recognition at the exterior surface of a zwitterionic telomer brush. *Langmuir* 2010;26:6767–74.
- [105] Ji Y, Lv R, Song S, Huang J, Zhang L, Huang G, et al. Facile fabrication of zwitterionic magnetic composites by one-step distillation-precipitation polymerization for highly specific enrichment of glycopeptides. *Anal Chim Acta* 2019;1053:43–53.
- [106] Liu CH, Faria AF, Ma J, Elimelech M. Mitigation of biofilm development on thin-film composite membranes functionalized with zwitterionic polymers and silver nanoparticles. *Environ Sci Technol* 2017;51:182–91.
- [107] Ye G, Lee J, Perreault F, Elimelech M. Controlled architecture of dual-functional block copolymer brushes on thin-film composite membranes for integrated “defending” and “attacking” strategies against biofouling. *ACS Appl Mater Interfaces* 2015;7:23069–79.
- [108] Leigh BL, Cheng E, Xu LJ, Andresen C, Hansen MR, Guyon CA. Photopolymerizable zwitterionic polymer patterns control cell adhesion and guide neural growth. *Biomacromolecules* 2017;18:2389–401.
- [109] Leng C, Han XF, Shao Q, Zhu YH, Li YT, Jiang SY, et al. In situ probing of the surface hydration of zwitterionic polymer brushes: structural and environmental effects. *J Phys Chem C* 2014;118:15840–5.
- [110] Han XF, Leng C, Shao Q, Jiang SY, Chen Z. Absolute orientations of water molecules at zwitterionic polymer interfaces and interfacial dynamics after salt exposure. *Langmuir* 2019;35:1327–34.
- [111] Wang T, Wang X, Long Y, Liu G, Zhang G. Ion-specific conformational behavior of polyzwitterionic brushes: exploiting it for protein adsorption/desorption control. *Langmuir* 2013;29:6588–96.
- [112] Zhang XY, Xie M, Yang Z, Wu HC, Fang CJ, Bai LM, et al. Antifouling double-skinned forward osmosis membranes by constructing zwitterionic brush-decorated MWCNT ultrathin films. *ACS Appl Mater Interfaces* 2019;11:19462–71.
- [113] Chou YN, Wen TC, Chang Y. Zwitterionic surface grafting of epoxylated sulfobetaine copolymers for the development of stealth biomaterial interfaces. *Acta Biomater* 2016;40:78–91.
- [114] Du T, Li B, Wang X, Yu B, Pei X, Huck WTS, et al. Bio-Inspired renewable surface-initiated polymerization from permanently embedded initiators. *Angew Chem Int Ed* 2016;55:4260–4.
- [115] Li MZ, Li JH, Shao XS, Miao J, Wang JB, Zhang QQ, et al. Grafting zwitterionic brush on the surface of PVDF membrane using physisorbed free radical grafting technique. *J Membr Sci* 2012;405:141–8.
- [116] Chang Y, Shih Y-J, Lai C-J, Kung H-H, Jiang S. Blood-inert surfaces via ion-pair anchoring of zwitterionic copolymer brushes in human whole blood. *Adv Funct Mater* 2013;23:1100–10.
- [117] Hirasawa M, Tsutsumi-Arai C, Takakusaki K, Oya T, Fueki K, Wakabayashi N. Superhydrophilic co-polymer coatings on denture surfaces reduce *Candida albicans* adhesion—An in vitro study. *Arch Oral Biol* 2018;87:143–50.
- [118] Cho WK, Kong B, Choi IS. Highly efficient non-biofouling coating of zwitterionic polymers: poly((3-(methacryloylamino)propyl)-dimethyl(3-sulfo)propyl)ammonium hydroxide). *Langmuir* 2007;23:5678–82.
- [119] Wang LY, Li GJ, Lin YL, Zhang ZX, Chen ZF, Wu SQ. A strategy for constructing anti-adhesion surfaces based on interfacial thiol-ene photoclick chemistry between DOPA derivatives with a catechol anchor group and zwitterionic betaine macromolecules. *Polym Chem* 2016;7:4964–74.
- [120] Yu DP, Guo ZM, Shen AJ, Yan JY, Dong XF, Jin GW, et al. Synthesis and evaluation of sulfobetaine zwitterionic polymer bonded stationary phase. *Talanta* 2016;161:860–6.
- [121] Huang KT, Fang YL, Hsieh PS, Li CC, Dai NT, Huang CJ. Non-sticky and antimicrobial zwitterionic nanocomposite dressings for infected chronic wounds. *Biomater Sci* 2017;5:1072–81.
- [122] Xiao SW, Zhang YX, Shen MX, Chen F, Fan P, Zhong MQ, et al. Structural dependence of salt-responsive polyzwitterionic brushes with an anti-polyelectrolyte effect. *Langmuir* 2018;34:97–105.
- [123] Lazarin AA, Machado AL, Zamperini CA, Wady AF, Spolidorio DMP, Vergani CE. Effect of experimental photopolymerized coatings on the hydrophobicity of a denture base acrylic resin and on *Candida albicans* adhesion. *Arch Oral Biol* 2013;58:1–9.
- [124] Kondo T, Nomura K, Murou M, Gemmei-Ide M, Kitano H, Noguchi H, et al. Structure of water in the vicinity of a zwitterionic polymer brush as examined by sum frequency generation method. *Colloids Surf B Biointerfaces* 2012;100:126–32.
- [125] Xie Y, Chen S, Qian Y, Zhao W, Zhao C. Photo-responsive membrane surface: switching from bactericidal to bacteria-resistant property. *Mater Sci Eng C* 2018;84:52–9.
- [126] Cao Z, Yu Q, Xue H, Cheng G, Jiang S. Nanoparticles for drug delivery prepared from amphiphilic PLGA zwitterionic block copolymers with sharp contrast in polarity between two blocks. *Angew Chem Int Ed* 2010;49:3771–6.
- [127] Keefe AJ, Jiang S. Poly(zwitterionic)protein conjugates offer increased stability without sacrificing binding affinity or bioactivity. *Nat Chem* 2012;4:59–63.
- [128] Nishida M, Nakaji-Hirabayashi T, Kitano H, Saruwatari Y, Matsuoka K. Titanium alloy modified with anti-biofouling zwitterionic polymer to facilitate formation of bio-mineral layer. *Colloids Surf B: Biointerfaces* 2017;152:302–10.
- [129] Li LF, Nakaji-Hirabayashi T, Tokiwa K, Kitano H, Ohno K, Usui Y, et al. UV-patterning of anti-biofouling zwitterionic copolymer layer with an aromatic anchor group. *Macromol Mater Eng* 2017;302:1600374.
- [130] Baker SL, Munasinghe A, Kaupbayeva B, Kang NR, Certiat M, Murata H, et al. Transforming protein-polymer conjugate purification by tuning protein solubility. *Nat Commun* 2019;10:4718.
- [131] Zhou Q, Lei X, Li J, Yan B, Zhang Q. Antifouling, adsorption and reversible flux properties of zwitterionic grafted PVDF membrane prepared via physisorbed free radical polymerization. *Desalination* 2014;337:6–15.
- [132] Zhang L, Cao Z, Bai T, Carr L, Ella-Menye J-R, Irvin C, et al. Zwitterionic hydrogels implanted in mice resist the foreign-body reaction. *Nat Biotechnol* 2013;31:553–6.
- [133] Silles I, Andrieu-Brunsen A. Programming ionic pore accessibility in zwitterionic polymer modified nanopores. *Langmuir* 2018;34:807–16.
- [134] Wang J, Hui N. Zwitterionic poly(carboxybetaine) functionalized conducting polymer polyaniline nanowires for the electrochemical detection of carcinoembryonic antigen in undiluted blood serum. *Bioelectrochemistry* 2019;125:90–6.
- [135] Luo X, Xu M, Freeman C, James T, Davis JJ. Ultrasensitive label free electrical detection of insulin in neat blood serum. *Nat Chem* 2013;5:4129–34.
- [136] Lisalová H, Brynda E, Houska M, Vášová I, Mrkvová K, Song XC, et al. Ultralow-fouling behavior of biorecognition coatings based on carboxy-functional brushes of zwitterionic homo- and copolymers in blood plasma: functionalization matters. *Anal Chem* 2017;89:3524–31.
- [137] Jia GW, Cao ZQ, Xue H, Xu YS, Jiang SY. Novel zwitterionic-polymer-coated silica nanoparticles. *Langmuir* 2009;25:3196–9.
- [138] Rodriguez-Emmenegger C, Houska M, Alles AB, Brynda E. Surfaces resistant to fouling from biological fluids: towards bioactive surfaces for real applications. *Macromol Biosci* 2012;12:1413–22.
- [139] Weinman ST, Bass M, Pandit S, Herzberg M, Freger V, Husson SM. A switchable zwitterionic membrane surface chemistry for biofouling control. *J Membr Sci* 2018;548:490–501.
- [140] Li L, Nakaji-Hirabayashi T, Kitano H, Ohno K, Kishioka T, Usui Y. Gradation of proteins and cells attached to the surface of bio-inert zwitterionic polymer brush. *Colloids Surf B Biointerfaces* 2016;144:180–7.
- [141] Suzuki H, Murou M, Kitano H, Ohno K, Saruwatari Y. Silica particles coated with zwitterionic polymer brush: formation of colloidal crystals and anti-biofouling properties in aqueous medium. *Colloids Surf B Biointerfaces* 2011;84:111–6.
- [142] Chen H, Yang J, Xiao S, Hu R, Bhaway SM, Vogt BD, et al. Salt-responsive polyzwitterionic materials for surface regeneration between switchable fouling and antifouling properties. *Acta Biomater* 2016;40:62–9.
- [143] Yang JT, Chen H, Xiao SW, Shen MX, Chen F, Fan P, et al. Salt-responsive zwitterionic polymer brushes with tunable friction and antifouling properties. *Langmuir* 2015;31:9125–33.
- [144] Sun YL, Chen CL, Xu H, Lei K, Xu GZ, Zhao L, et al. Surface modification of silicon wafer by grafting zwitterionic polymers to improve its antifouling property. *Appl Surf Sci* 2017;419:642–9.
- [145] Ye Q, He B, Zhang Y, Zhang J, Liu S, Zhou F. Grafting robust thick zwitterionic polymer brushes via subsurface-initiated ring-opening metathesis polymerization for antimicrobial and anti-biofouling. *ACS Appl Mater Interfaces* 2019;11:39171–8.
- [146] Li G, Xue H, Gao C, Zhang F, Jiang S. Nonfouling polyampholytes from an ion-pair comonomer with biomimetic adhesive groups. *Macromolecules* 2010;43:14–6.
- [147] Yuan Y-Y, Mao C-Q, Du X-J, Du J-Z, Wang F, Wang J. Surface charge switchable nanoparticles based on zwitterionic polymer for enhanced drug delivery to tumor. *Adv Mater* 2012;24:5476–80.
- [148] Zheng J, Li M, Yao Y, Zhang X, Wang L. Zwitterionic carbon nanotube assisted thin-film nanocomposite membranes with excellent efficiency for separation of mono/divalent ions from brackish water. *J Mater Chem A* 2017;5:13730–9.
- [149] Zhao X, Qiu ZS, Zhang YJ, Zhong HY, Huang WA, Tang ZC. Zwitterionic polymer P (AM-DMC-AMPS) as a low-molecular-weight encapsulator in deepwater drilling fluid. *Appl Sciences-Basel* 2017;7:594.
- [150] Ma T, Su Y, Li Y, Zhang R, Liu Y, He M, et al. Fabrication of electro-neutral nanofiltration membranes at neutral pH with antifouling surface via interfacial polymerization from a novel zwitterionic amine monomer. *J Membr Sci* 2016;503:101–9.
- [151] Xie Y, Chen L, Zhang X, Chen S, Zhang M, Zhao W, et al. Integrating zwitterionic polymer and Ag nanoparticles on polymeric membrane surface to prepare antifouling and bactericidal surface via Schiff-based layer-by-layer assembly. *J Colloid Interface Sci* 2018;510:308–17.
- [152] Venault A, Yang H-S, Chiang Y-C, Lee B-S, Ruaan R-C, Chang Y. Bacterial resistance control on mineral surfaces of hydroxyapatite and human teeth via surface charge-driven antifouling coatings. *ACS Appl Mater Interfaces* 2014;6:3201–10.

- [153] He Q, Küller A, Grunze M, Li J. Fabrication of thermosensitive polymer nanopatterns through chemical lithography and atom transfer radical polymerization. *Langmuir* 2007;23:3981–7.
- [154] Li DX, He Q, Cui Y, Wang KW, Zhang XM, Li JB. Thermosensitive copolymer networks modify gold nanoparticles for nanocomposite entrapment. *Chem Eur J* 2007;13:2224–9.
- [155] Yang Y, Yan XH, Cui Y, He Q, Li DX, Wang AH, et al. Preparation of polymer-coated mesoporous silica nanoparticles used for cellular imaging by a "graft-from" method. *J Mater Chem* 2008;18:5731–7.
- [156] Li DX, Liu N, Gao YY, Lin WH, Li CF. Thermosensitive polymer stabilized core-shell AuNR@Ag nanostructures as "smart" recyclable catalyst. *J Nanopart Res* 2017;19:377.
- [157] Huang Y, Sasano T, Tsujii Y, Ohno K. Well-defined polymer-brush-coated rod-shaped particles: synthesis and formation of liquid crystals. *Macromolecules* 2016;49:8430–9.
- [158] Ohno K, K-m Koh, Tsujii Y, Fukuda T. Synthesis of gold nanoparticles coated with well-defined, high-density polymer brushes by surface-initiated living radical polymerization. *Macromolecules* 2002;35:8989–93.
- [159] Chen S, Xie Y, Xiao T, Zhao W, Li J, Zhao C. Tannic acid-inspired and post-crosslinking of zwitterionic polymer as a universal approach towards antifouling surface. *Chem Eng J* 2018;337:122–32.
- [160] Xia YQ, Adibnia V, Shan CC, Huang RL, Qi W, He ZM, et al. Synergy between zwitterionic polymers and hyaluronic acid enhances antifouling performance. *Langmuir* 2019;35:15535–42.
- [161] Banquy X, Burdynska J, Lee DW, Matyjaszewski K, Israelachvili J. Bioinspired bottle-brush polymer exhibits low friction and amontons-like behavior. *J Am Chem Soc* 2014;136:6199–202.
- [162] Faivre J, Shrestha BR, Burdynska J, Xie G, Moldovan F, Delair T, et al. Wear protection without surface modification using a synergistic mixture of molecular brushes and linear polymers. *ACS Nano* 2017;11:1762–9.
- [163] Faivre J, Shrestha BR, Xie G, Delair T, David L, Matyjaszewski K, et al. Unraveling the correlations between conformation, lubrication, and chemical stability of bottle-brush polymers at interfaces. *Biomacromolecules* 2017;18:4002–10.
- [164] Faivre J, Shrestha BR, Xie G, Olszewski M, Adibnia V, Moldovan F, et al. Intermolecular interactions between bottlebrush polymers boost the protection of surfaces against frictional wear. *Chem Mater* 2018;30:4140–9.
- [165] Shafi HZ, Matin A, Khan Z, Khalil A, Gleason KK. Surface modification of reverse osmosis membranes with zwitterionic coatings: a potential strategy for control of biofouling. *Surf Coat Technol* 2015;279:171–9.
- [166] Yang R, Xu J, Ozaydin-Ince G, Wong SY, Gleason KK. Surface-tethered zwitterionic ultrathin antifouling coatings on reverse osmosis membranes by initiated chemical vapor deposition. *Chem Mater* 2011;23:1263–72.
- [167] Wang YF, Shen J, Yuan J. Design of hemocompatible and antifouling PET sheets with synergistic zwitterionic surfaces. *J Colloid Interface Sci* 2016;480:205–17.
- [168] Sun JT, Yu ZQ, Hong CY, Pan CY. Biocompatible zwitterionic sulfobetaine copolymer-coated mesoporous silica nanoparticles for temperature-responsive drug release. *Macromol Rapid Commun* 2012;33:811–8.
- [169] Zhu YZ, Xie W, Zhang F, Xing TL, Jin J. Superhydrophilic in-situ-cross-linked zwitterionic polyelectrolyte/PVDF-blend membrane for highly efficient oil/water emulsion separation. *ACS Appl Mater Interfaces* 2017;9:9603–13.
- [170] Fang LF, Jeon S, Kakihana Y, Kakehi J, Zhu BK, Matsuyama H, et al. Improved antifouling properties of polyvinyl chloride blend membranes by novel phosphate based-zwitterionic polymer additive. *J Membr Sci* 2017;528:326–35.
- [171] Matyjaszewski K. Atom transfer radical polymerization (ATRP): current status and future perspectives. *Macromolecules* 2012;45:4015–39.
- [172] Ran J, Wu L, Zhang Z, Xu T. Atom transfer radical polymerization (ATRP): a versatile and forceful tool for functional membranes. *Prog Polym Sci* 2014;39:124–44.
- [173] Li DX, Cui Y, Wang KW, He Q, Yan XH, Li JB. Thermosensitive nanostructures comprising gold nanoparticles grafted with block copolymers. *Adv Funct Mater* 2007;17:3134–40.
- [174] Li DX, He Q, Cui Y, Li JB. Fabrication of pH-responsive nanocomposites of gold nanoparticles/poly(4-vinylpyridine). *Chem Mater* 2007;19:412–7.
- [175] Li DX, Jang YJ, Lee J, Lee JE, Kochuveedu ST, Kim DH. Grafting poly(4-vinylpyridine) onto gold nanorods toward functional plasmonic core-shell nanostructures. *J Mater Chem* 2011;21:16453–60.
- [176] Baker SL, Kaupbayeva B, Lathwal S, Das SR, Russell AJ, Matyjaszewski K. Atom transfer radical polymerization for biorelated hybrid materials. *Biomacromolecules* 2019;20:4272–98.
- [177] Chiang Y-C, Chang Y, Higuchi A, Chen WY, Ruaan RC. Sulfobetaine-grafted poly(vinylidene fluoride) ultrafiltration membranes exhibit excellent antifouling property. *J Membr Sci* 2009;339:151–9.
- [178] Durand-Gasselin C, Koerin R, Rieger J, Lequeux N, Sanson N. Colloidal stability of zwitterionic polymer-grafted gold nanoparticles in water. *J Colloid Interface Sci* 2014;434:188–94.
- [179] Mi Y-F, Zhao F-Y, Guo Y-S, Weng X-D, Ye C-C, An Q-F. Constructing zwitterionic surface of nanofiltration membrane for high flux and antifouling performance. *J Membr Sci* 2017;541:29–38.
- [180] Leng C, Hung H-C, Sun S, Wang D, Li Y, Jiang S, et al. Probing the surface hydration of nonfouling zwitterionic and PEG materials in contact with proteins. *ACS Appl Mater Interfaces* 2015;7:16881–8.
- [181] Das S, Banik M, Chen G, Sinha S, Mukherjee R. Polyelectrolyte brushes: theory, modelling, synthesis and applications. *Soft Matter* 2015;11:8550–83.
- [182] Xiao S, Ren B, Huang L, Shen M, Zhang Y, Zhong M, et al. Salt-responsive zwitterionic polymer brushes with anti-polyelectrolyte property. *Curr Opin Chem Eng* 2018;19:86–93.
- [183] Martínez JS, Kelly KD, Ghoussoub YE, Delgado JD, Keller TCS, Schlenoff JB. Cell resistant zwitterionic polyelectrolyte coating promotes bacterial attachment: an adhesion contradiction. *Biomater Sci* 2016;4:689–98.
- [184] Sangsuwan A, Kawasaki H, Matsumura Y, Iwasaki Y. Antimicrobial silver nanoclusters bearing biocompatible phosphorylcholine based zwitterionic protects. *Bioconjug Chem* 2016;27:2527–33.
- [185] Encinas N, Angulo M, Astorga C, Colilla M, Izquierdo-Barba I, Vallet-Regí M. Mixed-charge pseudo-zwitterionic mesoporous silica nanoparticles with low-fouling and reduced cell uptake properties. *Acta Biomater* 2019;84:317–27.
- [186] Song W, Li Z, Li Y, You H, Qi P, Liu F, et al. Facile sol-gel coating process for anti-biofouling modification of poly(vinylidene fluoride) microfiltration membrane based on novel zwitterionic organosilica. *J Membr Sci* 2018;550:266–77.
- [187] Qiu H, Wanigasekara E, Zhang Y, Tran T, Armstrong DW. Development and evaluation of new zwitterionic Hydrophilic interaction liquid chromatography stationary phases based on 3-P,P-diphenylphosphonium-propylsulfonate. *J Chromatogr A* 2011;1218:8075–82.
- [188] Lei H, Wang M, Tang Z, Luan Y, Liu W, Song B, et al. Control of lysozyme adsorption by pH on surfaces modified with polyampholyte brushes. *Langmuir* 2014;30:501–8.
- [189] Charnley M, Textor M, Acikgoz C. Designed polymer structures with antifouling-antimicrobial properties. *React Funct Polym* 2011;71:329–34.
- [190] Konno T, Ishihara K. Temporal and spatially controllable cell encapsulation using a water-soluble phospholipid polymer with phenylboronic acid moiety. *Biomaterials* 2007;28:1770–7.
- [191] Zhou L, Tan G-X, Ning C-Y. Modification of biomaterials surface by mimetic cell membrane to improve biocompatibility. *Front Mater Sci* 2014;8:325–31.
- [192] Seo J, Seo JH. Fabrication of an anti-biofouling plasma-filtration membrane by an electrospinning process using photo-cross-linkable zwitterionic phospholipid polymers. *ACS Appl Mater Interfaces* 2017;9:19591–600.
- [193] Wang S-Y, Fang L-F, Cheng L, Jeon S, Kato N, Matsuyama H. Novel ultrafiltration membranes with excellent antifouling properties and chlorine resistance using a poly(vinyl chloride)-based copolymer. *J Membr Sci* 2018;549:101–10.
- [194] Boersma PJ, Mohammed S, Heck AJR. Hydrophilic interaction liquid chromatography (HILIC) in proteomics. *Anal Bioanal Chem* 2008;391:151–9.
- [195] Jandera P, Staňková M, Hájek T. New zwitterionic polymethacrylate monolithic columns for one- and two-dimensional microliquid chromatography. *J Sep Sci* 2013;36:2430–40.
- [196] Zhang R, Watson DG, Wang L, Westrop GD, Coombs GH, Zhang T. Evaluation of mobile phase characteristics on three zwitterionic columns in hydrophilic interaction liquid chromatography mode for liquid chromatography-high resolution mass spectrometry based untargeted metabolite profiling of Leishmania parasites. *J Chromatogr A* 2014;1362:168–79.
- [197] Guerrasio R, Haberhauer-Troyer C, Mattanovich D, Koellensperger G, Hann S. Metabolic profiling of amino acids in cellular samples via zwitterionic sub-2 μ m particle size HILIC-MS/MS and a uniformly ^{13}C labeled internal standard. *Anal Bioanal Chem* 2014;406:915–22.
- [198] Contrepas K, Jiang L, Snyder M. Optimized analytical procedures for the untargeted metabolomic profiling of human urine and plasma by combining hydrophilic interaction (HILIC) and reverse-phase liquid chromatography (RPLC)-mass spectrometry. *Mol Cell Proteomics* 2015;14:1684–95.
- [199] Wernisch S, Pennathur S. Evaluation of coverage, retention patterns, and selectivity of seven liquid chromatographic methods for metabolomics. *Anal Bioanal Chem* 2016;408:6079–91.
- [200] Elmsjö A, Haglöf J, Engskog MKR, Erngren I, Nestor M, Arvidsson T, et al. Method selectivity evaluation using the co-feature ratio in LC/MS metabolomics: comparison of HILIC stationary phase performance for the analysis of plasma, urine and cell extracts. *J Chromatogr A* 2018;1568:49–56.
- [201] Jandera P, Hájek T, Šromová Z. Comprehensive two-dimensional monolithic liquid chromatography of polar compounds. *J Sep Sci* 2019;42:670–7.
- [202] Jandera P, Hájek T, Šromová Z. Mobile phase effects in reversed-phase and hydrophilic interaction liquid chromatography revisited. *J Chromatogr A* 2018;1543:48–57.
- [203] Aggarwal P, Hall JB, Mcleland CB, Dobrovolskaia MA, Mcneil SE. Nanoparticle interaction with plasma proteins as it relates to particle biodistribution, biocompatibility and therapeutic efficacy. *Adv Drug Deliv Rev* 2009;61:428–37.
- [204] Cao Z, Jiang S. Super-hydrophilic zwitterionic poly(carboxybetaine) and amphiphilic non-ionic poly(ethylene glycol) for stealth nanoparticles. *Nano Today* 2012;7:404–13.
- [205] García KP, Zarschler K, Barbaro L, Barreto JA, O'Malley W, Spiccia L, et al. Zwitterionic-coated "Stealth" nanoparticles for biomedical applications: recent advances in countering biomolecular corona formation and uptake by the mononuclear phagocyte system. *Small* 2014;10:2516–29.
- [206] Xiao W, Lin J, Li M, Ma Y, Chen Y, Zhang C, et al. Prolonged in vivo circulation time by zwitterionic modification of magnetite nanoparticles for blood pool contrast agents. *Contrast Media Mol Imaging* 2012;7:320–7.